

Chapter 13

Digital Television Channel Coding and Modulation

13.1. INTRODUCTION

Although the MPEG-2 standards specify source coding and multiplexing for digital television, they do not specify the channel coding and modulation to be used to transmit the systems bit stream over any particular channel. This chapter describes the channel coding and modulation provided for terrestrial broadcast by the DVB and ATSC digital television standards.¹ Generic concepts of channel coding and modulation, common to both standards are described first, followed by more detailed description of the methods employed in DVB [1] and ATSC. The sections relating to DVB and ATSC are written so that each can be read independent of the other.

The aim of this chapter is to provide an overview of the modulation techniques used in digital television systems. Some familiarity with digital modulation is therefore assumed. A comprehensive coverage of digital modulation and channel coding would require a book in its own right.

13.2. GENERIC CONCEPTS

Digital signals passing through transmission channels are subject to errors introduced in the transmission process. Although all channels introduce errors, some channels, especially radio channels, tend to introduce high error rates. The types of impairments introduced by an imperfect transmission channel include additive noise and channel perturbations. Additive noise may take the form of Gaussian noise with stationary statistics, impulsive noise that is not always stationary or easy to characterize, or even deliberate jamming. Channel perturbations may occur due to fading in radio channels, synchronization slip in digital channels, and breaks in transmission.

¹We do not discuss satellite or cable delivery systems in this chapter. Information on these systems can be found in the relevant standards.

The likely effects of these channel impairments on a digital signal are one of *uniformly random errors*—errors occurring individually and independently, with approximately uniform probability density, primarily due to noise (often just called *random noise*); *burst errors*—errors grouped in clusters, mainly the result of a combination of noise and channel perturbations; and *erasures*—irregular intervals when it is known that no reliable signal can be detected because of severe channel perturbation.

Minimization of the impact of transmission errors involves a combination of channel coding and modulation, each of which are addressed in the following sections.

13.2.1. Channel Characteristics and Intersymbol Interference

Transfer of information across a communications channel is limited by the bandwidth of the channel and the combination of its bandwidth and signal-to-noise ratio.

For a channel that introduces *additive, white, Gaussian noise* (AWGN), the precise relationship between channel capacity, bandwidth, and signal-to-noise ratio is given by Shannon's criterion:

$$C = B \log_2 \left(1 + \frac{S}{N} \right)$$

where C is the capacity of the channel in bits per second, B is the bandwidth of the channel in hertz, S is the signal power at the receiver, and N is the noise power at the receiver [2]. In practice, the throughput of a channel is usually significantly less than the maximum indicated by the value of C . The major reason for this is limitations in the performance of channel coding systems. Shannon's criterion tells us that an increase in channel capacity can be achieved either by increasing its bandwidth or increasing its signal-to-noise ratio. It also tells us that channel bandwidth has a much greater impact than its signal-to-noise ratio.

Limited channel bandwidth can blur transitions where changes of state occur in a bit stream. This effect, known as intersymbol interference, is illustrated in Figure 13.1. An equalizer is a digital filter that has been designed to correct for

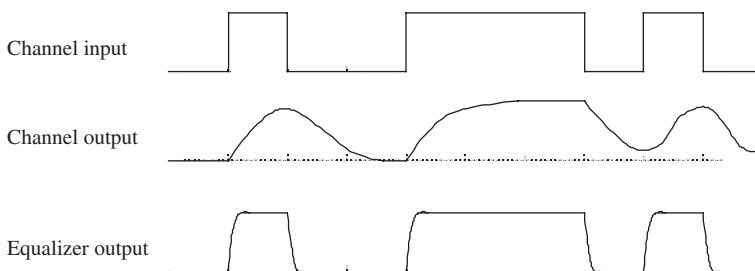


Figure 13.1 Intersymbol interference and equalization for a binary waveform.

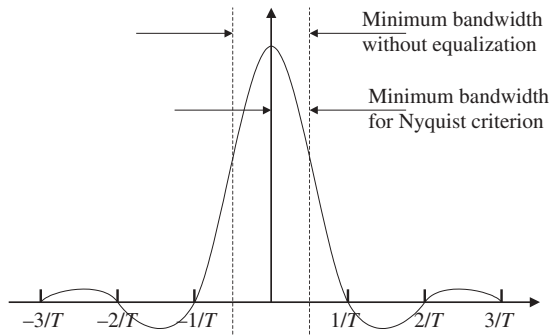


Figure 13.2 Minimum channel bandwidths required to meet the Nyquist criterion and to avoid the need for equalization.

the intersymbol interference of the channel. The use of an appropriately designed *equalizer* can allow the original signal to be recovered so long as the channel bandwidth meets the Nyquist criterion, which is discussed below.

Independent of any limitation on information-carrying capacity implied by Shannon's criterion, the bandwidth of a channel limits the symbol rate that can be carried. The Nyquist criterion says that the minimum bandwidth required to carry a given symbol rate is equal to half the symbol rate [3]. Figure 13.2 illustrates the minimum bandwidth requirements with and without equalization for a signal with symbol period T .

EXAMPLE 13.1—Nyquist Criterion

A bit stream is transmitted at a rate of 120,000 bit/s, with each bit carried in one symbol. What is the minimum channel bandwidth required to meet the Nyquist criterion? What is the impact if each symbol carries two bits of information?

The minimum channel bandwidth to meet the Nyquist criterion is 60 kHz. If the same bit stream is transmitted with two bits in each transmitted symbol (that is, at 60,000 sym/s), then the minimum bandwidth to meet the Nyquist criterion is 30 kHz. ■

As the data rate rises or channel bandwidth falls, the impact of intersymbol interference increases. If significant intersymbol interference occurs, a receiver must incorporate an equalizer to recover the original bit stream.

In this section the following notation is used:

- $c(t)$ —the impulse response of channel;
- $C(f)$ —the frequency response of channel, which is the Fourier transform of the impulse response;
- W —the bandwidth of bandlimited channel;
- $\theta(f)$ —the phase component of $C(f)$; and
- $\tau(f)$ —the delay response $= -(1/2\pi) d\theta(f)/df$.

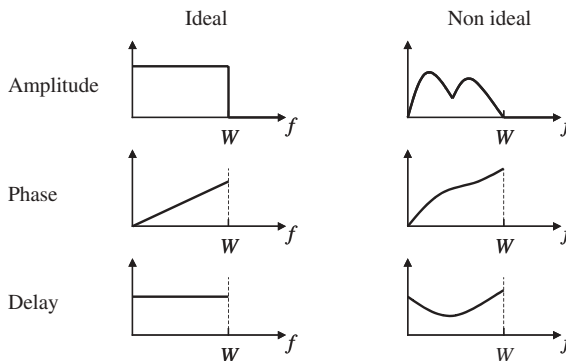


Figure 13.3 Ideal versus nonideal channels.

The characteristics of ideal and nonideal channels are illustrated in Figure 13.3. An ideal channel is one with the following characteristics:

- *Constant in-band amplitude response.* $|C(f)|$ is constant for $|f| \leq W$. Violation $\Rightarrow$ “amplitude distortion.”
- *Zero out-of-band amplitude response.* $C(f) = 0$ for $|f| > W$.
- *Phase linearity.* $\theta(f)$ is a linear function of f for $|f| \leq W$. Violation $\Rightarrow$ “phase distortion.”
- *Constant delay.* $\tau(f)$ is constant for $|f| \leq W$. Violation $\Rightarrow$ “delay distortion.”

13.2.2. Modulation

A digital bit stream is a representation of data in terms of the binary values 0 and 1. Inside a video coder or decoder, these values are often represented as voltage levels. They cannot normally be transmitted over a communications system directly in this form. The purpose of modulation is to convert the binary values into a form that is suitable for transmission. Currently used transmission systems usually represent the information content of the bit stream in the amplitude, the frequency, or the phase of the transmitted signal. For digital signals, these forms of modulation are referred to as *amplitude-shift keying (ASK)*, *frequency-shift keying (FSK)*, and *phase-shift keying (PSK)*.

The forms of modulation used for digital television are PSK or combinations of ASK and PSK. The simplest form of modulation used is binary PSK (BPSK). The waveforms used by BPSK to represent the binary values 0 and 1 are illustrated on the left side of Figure 13.4. For each bit entering the modulator, one symbol is produced at the output. Each BPSK symbol, therefore, carries one bit of data. These waveforms are simply $\pm \cos$ functions. An alternative representation, in terms of the phase of waveforms, is shown on the right side of Figure 13.4. BPSK offers a high level of robustness in the presence of additive channel noise (significantly higher than, e.g., ASK) because the information is contained in the phase of the

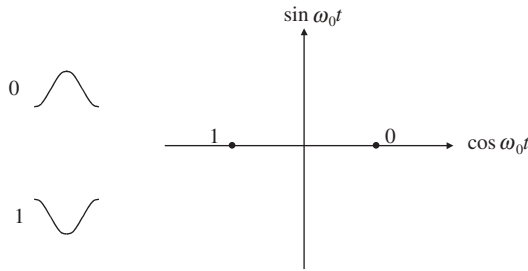


Figure 13.4 Constellation diagram for BPSK.

transmitted signal, which is not directly affected by the additive noise, rather than in the amplitude, which is directly affected by additive noise.

Differential binary PSK, often denoted by DPSK, represents a binary 1 by a change of phase of π radians (one half cycle) and a binary 0 by no change of phase.

In quadrature PSK (QPSK), the number of symbols is increased from two to four, producing the constellation diagram shown in Figure 13.5. One QPSK symbol is produced at the output of the modulator for each pair of bits at its input. Each QPSK symbol, therefore, carries two bits of data. QPSK transports data at twice the rate as BPSK in the same channel bandwidth and offers identical error performance to BPSK with the same energy per bit on the transmission channel. This is possible because the QPSK waveform is made up of two orthogonal components: the in-phase (I) component, shown as the horizontal (real) axis in Figure 13.5, and the quadrature (Q) component, shown as the vertical (imaginary) axis. The cost of increase in channel capacity depends on a small increase in encoder and decoder complexity. For encoders and decoders implemented within modern integrated circuits, this additional cost is negligible.

The data carrying capacity of a channel can be increased beyond that available using QPSK without increasing the channel bandwidth, by increasing the number of bits of data carried by each modulation symbol. This requires increasing the number of points in the constellation diagram, which implies a reduction in the spacing between these points for a fixed transmission power. For digital television applications, this increase is achieved by making use of both the amplitude and phase of the

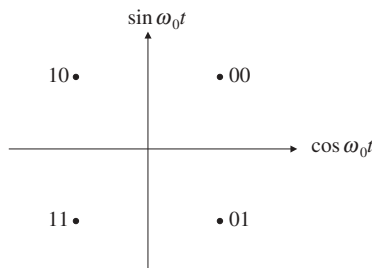


Figure 13.5 Constellation diagram for QPSK.

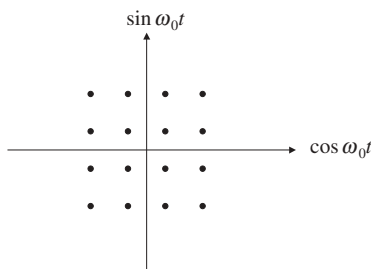


Figure 13.6 Constellation diagram for 16-QAM.

transmitted signal to carry information. The cost of this increase in channel capacity depends both on the complexity of encoders and decoders (which is acceptable with modern integrated circuits) and on a reduced robustness to additive noise introduced in the channel, resulting in an increased error rate.

The use of this combination of amplitude and phase modulation to carry four bits per symbol results in the 16-symbol constellation diagram of 16-quadrature amplitude modulation (16-QAM) as shown in Figure 13.6. Further increasing the number of bits per symbol to 6 leads to the constellation diagram of 64-QAM (Fig. 13.7). The further increase in data rate going from 16-QAM to 64-QAM comes with a corresponding reduction in robustness to channel noise. Using 16-QAM, a channel bandwidth of at least 3 kHz is required to carry a bit stream with rate 12 kbit/s without equalization, which reduces to 2 kHz for 64-QAM.

The amplitude spectrum of QPSK, 16-QAM, and 64-QAM is shown in Figure 13.8. The width of the central lobe is $2/T$, where T is the symbol period.

An alternative strategy is the combination of ASK with QPSK, which is used to combine ASK with BPSK. An example of this approach is the 8-level ASK modulation scheme illustrated in Figure 13.9. There are eight possible symbols, each of which carries three bits of data. These are illustrated on the left side of Figure 13.9, with the corresponding constellation diagram on the right side. The spectrum of this form of modulation is the same as illustrated in Figure 13.8, with T being the symbol

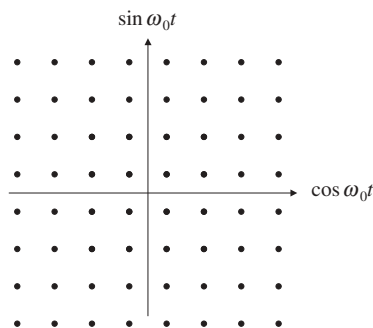


Figure 13.7 Constellation diagram for 64-QAM.

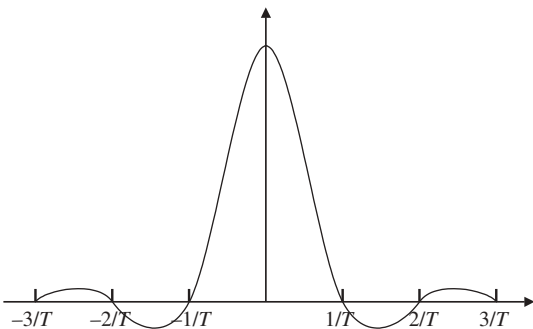


Figure 13.8 Spectrum of QPSK, 16-QAM, and 64-QAM.

period. Using this 8-level ASK, a channel bandwidth of at least 4 kHz is required to carry a bit stream with rate 12 kbit/s without significant intersymbol interference.

This 8-level ASK waveform may be filtered to reduce the transmission bandwidth. Theoretically, the minimum bandwidth that can be achieved without loss of information is the Nyquist bandwidth. This can be achieved by retaining the upper sideband and removing the lower sideband by filtering. In practice, a small amount of the lower sideband is usually retained, giving vestigial sideband modulation.

BPSK and QPSK have identical error performance because QPSK is formed from an orthogonal combination of two BPSK signals. In the same way, 64-QAM can be formed by the combination of two of the 8-level ASK modulation illustrated in Figure 13.9. The 8-level ASK modulation scheme should, therefore, have the same error performance as 64-QAM.

The bandwidth efficiency of this 8-level ASK scheme can be maximized by filtering, as illustrated in Figure 13.10, to limit the total bandwidth to the Nyquist

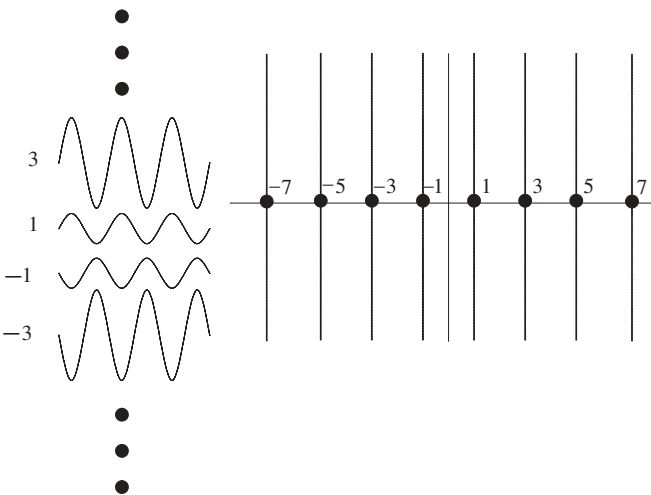


Figure 13.9 Symbol shapes and constellation diagram for 8-level ASK.

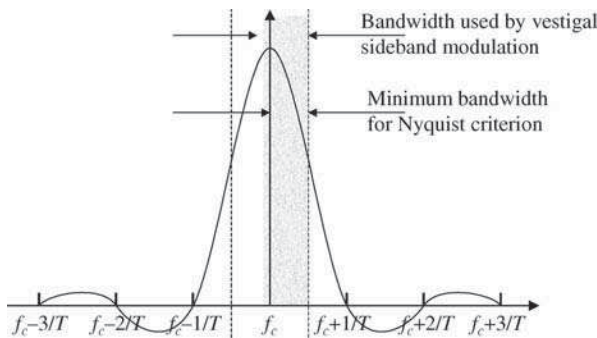


Figure 13.10 Frequency components used in 8-level vestigial sideband (8-VSB) modulation with carrier frequency f_c .

bandwidth. T is the period of a single 8-level symbol. Because the power density of the signal at the boundary between upper and lower sidebands is nonzero, the use of upper-sideband or lower-sideband transmission is not possible. Vestigial sideband modulation, in which the upper sideband and a small proportion of the lower sideband are retained, is preferred.

13.2.3. Equalization

Usually, an equalizer is an integral part of the demodulator. In the general case, modeling of the impact of channel and equalizer is quite difficult. Given the restrictions, however, that the channel noise is additive, white Gaussian and the channel is time invariant, a linear transversal model can be used for the combined impact of modulator, channel, and demodulator, that is, their combined effect can be modeled as an IIR filter [4]. This is illustrated in Figure 13.11.

13.2.3.1. Fixed-Tap Equalizer

In situations where the channel response is known in advance and remains constant, an optimal equalizer can be designed off-line. The equalizer consists of two parts:

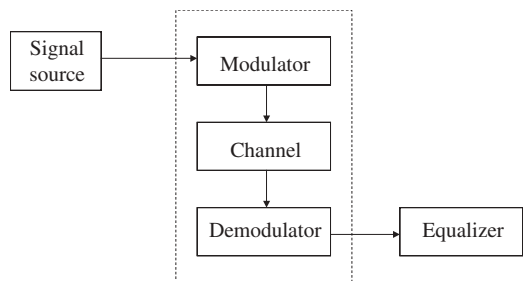


Figure 13.11 Linear-transversal model for channel.

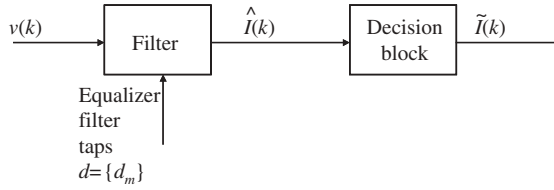


Figure 13.12 Fixed-tap equalizer.

an FIR filter and a decision block (Fig. 13.12). The filter approximately reverses the intersymbol interference introduced by the channel. The filter block is usually an FIR linear transversal filter, with taps denoted by the M -element vector $\{d_m\}$. The decision block takes the output of the filter and selects the most appropriate symbol as the output of the equalizer. The channel input at time k is denoted by $I(k)$, whereas the channel output at time k is denoted by $v(k)$. The equalizer output is an estimate of the channel input, so the equalizer output at time k is denoted by $\tilde{I}(k)$, so

$$\hat{I}(k) = \sum_{m=0}^{M-1} d_m \cdot v(k-m)$$

13.2.3.2. Trained Adaptive Equalizer

In a trained adaptive equalizer, the optimal values for the filter taps are estimated using knowledge of the original value of the incoming bit stream. This information is available in systems, such as ATSC, where a known training sequence is transmitted at regular intervals.

The operation of the filter and decision blocks is the same as that for a fixed equalizer (Fig. 13.13).

The adaptation block calculates updates to the equalizer filter coefficients based on the equalizer outputs and original symbol values. Because the optimal value of the equalizer filter taps is now estimated rather than known *a priori*, it is usual to denote the values of the taps by $\hat{d}(k)$, with the m th tap of the equalizer denoted by $\hat{d}_m(k)$.

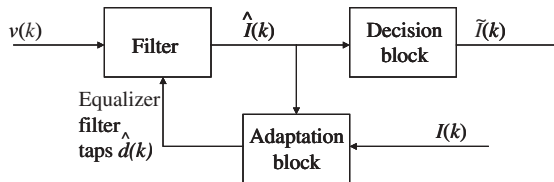


Figure 13.13 Trained adaptive equalizer.

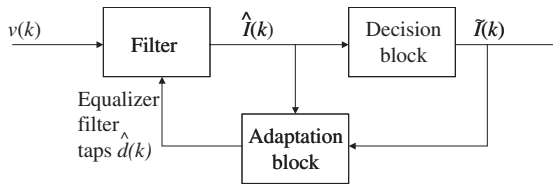


Figure 13.14 Blind adaptive equalizer.

13.2.3.3. Blind Adaptive Equalizer

In a blind-adaptive equalizer, the optimal values for the filter taps are estimated without access to the original value of the incoming bit stream. Instead, estimates of these symbol values based on equalizer output are used. Blind-adaptive equalizers do not require training sequences or explicit knowledge of channel characteristics.

The operation of the filter and decision blocks is identical to that discussed above for trained equalizers. The adaptation block calculates updates to the equalizer filter coefficients based on the equalizer outputs and decision block outputs (Fig. 13.14).

13.2.3.4. The LMS Algorithm

The least mean square (LMS) algorithm is one of the most fundamental techniques used in adaptive equalization and adaptive control. The LMS algorithm updates equalizer taps $d(k)$ in accordance with

$$\hat{d}_m(k) = \hat{d}_m(k-1) + \Delta \varepsilon(k) v^*(k-m)$$

where Δ is known as the step size, and where the error is given by

$$\varepsilon(k) = I(k) - \hat{I}(k)$$

LMS is a gradient descent algorithm. This means that the equalizer uses its measurement of the error to determine the magnitude and direction of the update at each step using a notational performance surface in which the best performance lies at the bottom of a well.

Convergence of the LMS algorithm is guaranteed if Δ is small enough. Increasing Δ has two effects:

- the rate of convergence increases and
- the long-term error is increased.

13.2.3.5. Practical Blind Adaptive Equalizers

Blind LMS In cases where a training sequence is not available, the LMS algorithm can be used for blind adaptation by replacing the training sequence

input to the adaptation process by the equalizer's output, using the same notation as in the previous section

$$\hat{d}_m(k) = \hat{d}_m(k-1) + \Delta \varepsilon(k) v^*(k-m)$$

where

$$\varepsilon(k) = \tilde{I}(k) - \hat{I}(k)$$

This algorithm has many of the same properties as the LMS algorithm. Its convergence, however, cannot be guaranteed. It is only suitable for situations where a reasonable approximation to the channel response is known *a priori*. This approach might be used to continue adaptation throughout a long transmission where a training sequence is used to provide the initial adaptation at the beginning of the transmission.

Godard's Algorithm Another method for blind equalization is that proposed by Godard [5]. In this algorithm, equalizer taps are updated as follows:

$$\hat{d}_{mk}(k) = d_m(k-1) + \Delta v^*(k-m) \hat{I}(k) \left(R_2 - |\hat{I}(k)|^2 \right)$$

where

$$R_2 = \frac{E(|I(k)|^4)}{E(|I(k)|^2)}$$

For modulation techniques whose alphabets include symbols with nonzero imaginary parts, all calculations in the equalizer are carried out on complex numbers (Table 13.1).

Convergence of the Godard algorithm can be guaranteed if the symbol stream is pseudorandom, an equalizer of infinite length is used, and the equalizer is initialized with all its taps set to zero, except for the center tap that is set to one.

13.2.4. Randomization

Before the demodulator in a receiver can recover a bit stream from the received signal, it must extract timing information. As a minimum, this timing information is used to identify the boundaries between symbols. A demodulator often relies on changes in the value of received symbols to identify these boundaries.

Figure 13.15 illustrates a BPSK received signal with and without transitions in the symbol value. In the former case, the demodulator has no means of identifying

Table 13.1 Alphabets and R_2 values for various modulation techniques using the Godard equalizer.

Modulation	Alphabet	R_2
BPSK	$\{1, -1\}$	1
QPSK	$\{1+j, 1-j, -1+j, -1-j\}$	2
16-QAM	$\{3+3j, 3+j, 1+3j, 1+j, -3+3j, -3+j, -1+3j, -1+j, 3-3j, 3-j, 1-3j, 1-j, -3-3j, -3-j, -1-3j, -1-j\}$	13.2
64-QAM	$\{7+7j, 7+5j, 5+7j, 5+5j, 7+j, 7+3j, 5+j, 5+3j, 1+7j, 1+5j, 3+5j, 3+7j, 1+j, 3+j, 1+3j, 3+3j, -7+7j, -7+5j, -5+7j, -5+5j, -7+j, -7+3j, -5+j, -5+3j, -1+7j, -1+5j, -3+5j, -3+7j, -1+j, -3+j, -1+3j, -3+3j, 7-7j, 7-5j, 5-7j, 5-5j, 7-j, 7-3j, 5-j, 5-3j, 1-7j, 1-5j, 3-5j, 3-7j, 1-j, 3-j, 1-3j, 3-3j, -7-7j, -7-5j, -5-7j, -5-5j, -7-j, -7-3j, -5-j, -5-3j, -1-7j, -1-5j, -3-5j, -3-7j, -1-j, -3-j, -1-3j, -3-3j\}$	58
8-VSB	$\{7, 5, 3, 1, -1, -3, -5, -7\}$	37

where boundaries between symbols lie. In the latter case, the presence of transitions between different symbol values makes this possible.

Long runs of identical symbols may often occur in digital television. Examples include start codes used in video and systems layers and transport stream null packets. In order to prevent these runs of identical symbols, a process of *transport stream randomization* is applied by both DVB and ATSC to the transport stream prior to channel coding. This is illustrated in Figure 13.16. The encoder generates a pseudorandom sequence. The exclusive-or of each bit of this pseudorandom sequence with the transport stream generated by the MPEG-2 systems encoder is passed through to the channel coding.

The decoder generates the same pseudorandom sequence as the encoder. The exclusive-or of this sequence with the received stream recovers the original bit

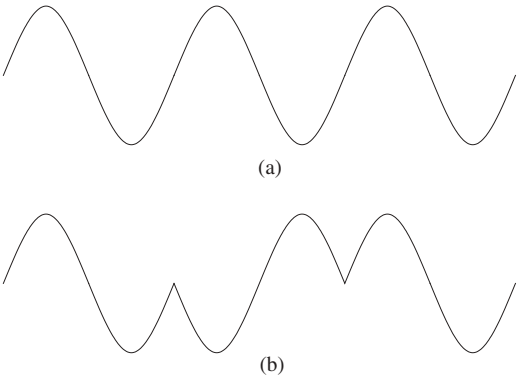


Figure 13.15 Received BPSK signal (a) without transitions and (b) with transitions in symbol value.

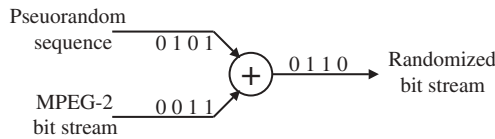


Figure 13.16 Transport stream randomization.

stream. To facilitate decoder operation, both DVB and ATSC reset the state of the pseudorandom sequence generator periodically.

Transport stream randomization does not guarantee that long runs of identical symbols cannot occur; rather it makes these events very unlikely. Although a similar effect could be achieved by the use of a periodic sequence, long streams of identical symbols would then be converted into long periodic streams. These long periodic streams may also upset the operation of a receiver.

13.2.5. Channel Coding Technology

Channel coding is used to correct errors caused by channel impairments through the introduction of controlled redundancy, enabling messages corrupted in transmission to be corrected before further processing [6]. With this controlled redundancy, only a subset of all possible transmitted messages (bit sequences) contains valid messages. This subset is called a code and the valid messages are called *code words* or *code vectors*. A good code is one in which code words are so separated that the likelihood of errors corrupting one into another is kept small.

Error detection is simplified to answering this question: Is the received message a code word or not? If it is a code word, one assumes that no errors have occurred. The probability of an undetected error getting through is then the probability of sufficient errors occurring to transform the real transmitted code word into another, apparently correct but in reality a false one.

If an error is detected, it can be corrected (at least in principle) by *automatic repeat request (ARQ)* or *forward error correction (FEC)*. In ARQ systems, the receiver checks received blocks of data for errors. If an error is detected, the receiver sends a request for retransmission to the transmitter. For broadcast services, such as digital television, ARQ is not feasible because it is not possible for a receiver to request retransmission. FEC is, therefore, used to protect transmitted data.

In FEC, the recipient corrects the errors by finding the valid code word ‘nearest’ to the received message, on the assumption that the nearest is the most likely because few corrupting errors are more likely than many. There are two types of FEC: *block coding* and *convolutional coding*.

In block coding, source data is partitioned into blocks of k bits, converted by the encoder into blocks of n ($>k$) bits with enough checks to enable the decoder to correct errors of the more probable kinds. The most common types of block codes [7] are Golay Codes, Bose–Chadhuri–Hocquenghem (BCH) Codes, and Reed–Solomon (RS) Codes. Block codes are used when information is naturally structured in blocks,

when channel capacity is relatively low and we do not want to waste it further with unnecessarily low code rates, and when quick efficient decoding is required because of limited processing time available.

For a convolutional code, the encoder operates not on disjoint blocks but on a running block of bits held in a shift register, generating a sequence of higher rate. The correcting capabilities of convolutional codes are not so clear-cut as with block codes. Probabilistic decoding, approximating maximum likelihood, is generally used. When long streams of relatively unstructured data are transmitted on high-capacity channels (such as terrestrial broadcast channels with capacities of approximately 20 Mbit/s) and when the complexity of the decoder represents a relatively small proportion of the total cost of the receiving equipment (such as a digital television set-top box), then convolutional codes can offer the best error-correcting solutions.

13.2.5.1. Block Codes

FEC allows transmission errors to be corrected by adding redundancy to a bit stream. This may seem to contradict the aim of source coding, which is to minimize the number of bits to be transmitted. The key difference between the original source redundancy and that added for FEC is that FEC adds (usually small amounts of) controlled redundancy in such a way as to maximize error correction.

A very simple example of a block code would be to represent each bit of data with a 5-bit symbol: $1 \rightarrow 11111$ and $0 \rightarrow 00000$. Using this code, two errors in each code word can be detected and corrected. If, for example, the transmitted code word is 11111 and the received code word 01110, the receiver converts this code word to a 1 because its value is closer to 11111 than to 00000. The obvious drawback of this very simple approach is that it has a very high overhead (400%).

Practical block codes group symbols into blocks and provide parity symbols for each block. Each symbol may consist of one or more bits, with the number of bits per symbol fixed for all symbols in the block. This greatly reduces the overhead, although at the cost of some reduction in performance. This is illustrated in Figure 13.17. This code with k information symbols and $n-k$ parity symbols is referred to as an (n, k) code.

If all valid code words differ in at least $d_{\min}$ symbols, the number of symbol errors t that the code is capable of correcting is

$$t = \left\lfloor \frac{d_{\min} - 1}{2} \right\rfloor$$

where $\lfloor x \rfloor$ denotes the largest integer whose value is not greater than x .

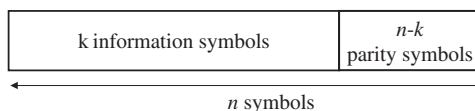


Figure 13.17 Structure of a practical block code.

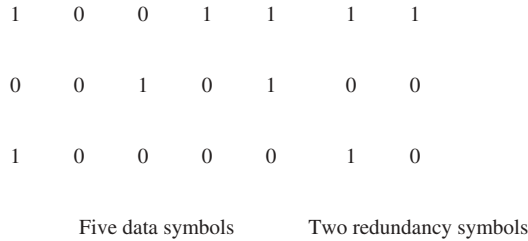


Figure 13.18 Example of block code using (7,5) RS code.

In the simple example above, each bit is one symbol, $d_{\min}$ is 5, so $t = 2$, meaning that two bit errors can be corrected.

The block codes used in digital television transmission are Reed–Solomon (RS) codes. RS codes use a nonbinary alphabet, that is, each symbol consists of more than one bit. One code word of an (7,5) RS code is illustrated in Figure 13.18. Each symbol contains three bits. Seven symbols make up one code word. The first five of these symbols contain the original data; the other two provide the redundancy for error correction.

This code can correct one symbol error, where a symbol error is defined as any number of bit errors in a single symbol. Reed–Solomon codes are therefore good for correcting burst errors, because there is no additional cost for correcting two errors in one symbol compared to a single error. In this example, the code could correct up to three bit errors provided they are all in the same symbol.

For k -bit symbols, the maximum code word length (n) for a Reed–Solomon code is $2k-1$. Typically 8-bit symbols are used, which means that each code word can contain up to 255 bytes.

A Reed–Solomon code that can correct t symbol errors requires $2t$ parity symbols. A Reed–Solomon code that can correct t symbol errors can also correct $2t$ erasures (i.e., where position of errors are known). In general, correction is possible if

$$2s + r < 2t$$

where s is the number of symbol errors and r is the number of erasures.

Reed–Solomon codes can be shortened by setting a number of symbols to zero at the encoder, transmitting the code word without these zero symbols and then reinserting these zero symbols at the decoder. For example, a (255,223) code can correct 16 symbol errors. A (200,168) code could operate by adding 55 zero bytes to create a (255,223) code and then transmitting only 168 data bytes plus 32 parity bytes. The notation (200,168,8) is often used to represent an RS (200,168) coder that operates on symbols of 8 bits.

The performance of RS codes is illustrated in Figure 13.19. The error rate at the input to the encoder is expressed as the symbol error rate on the horizontal axis. The residual error rate at the output of the coder (expressed as a bit error rate) is shown on the vertical axis.

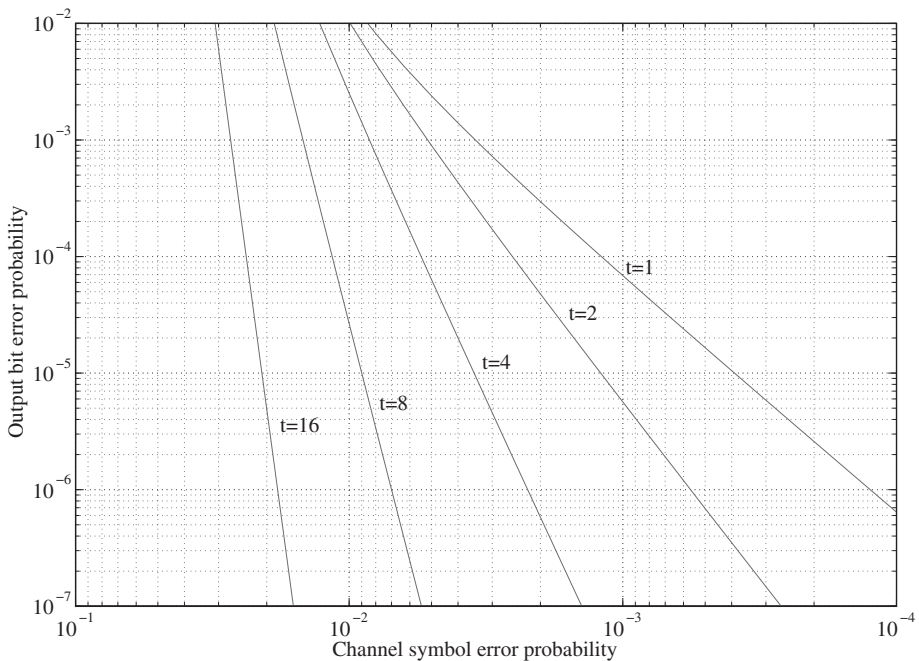


Figure 13.19 Performance of Reed–Solomon codes.

One disadvantage of block codes is that their performance depends on the location of errors in the bit stream. In the top bit stream illustrated in Figure 13.20, a particular pattern of three errors is split across the boundary of two blocks, with two errors occurring in the first block and one in the second. In the bottom bit stream, all three errors occur in the same block. The use of a block code capable of correcting two errors leads to an error-free output bit stream in the top case, but not in the bottom case.

13.2.5.2. Convolutional Codes

Convolutional codes operate on continuous streams of data and are commonly used on high-capacity wireless communications channels such as terrestrial broadcast

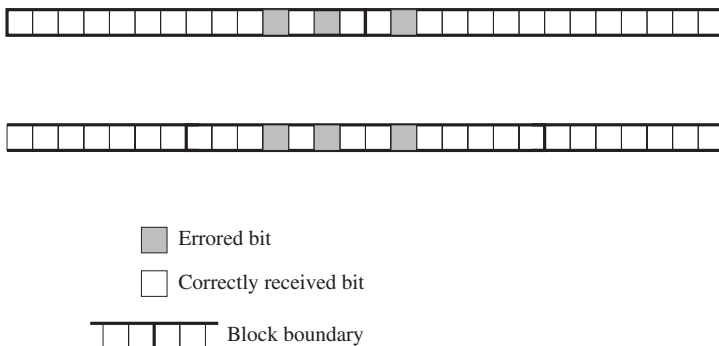


Figure 13.20 Impact of location of block boundaries on the performance of block codes.

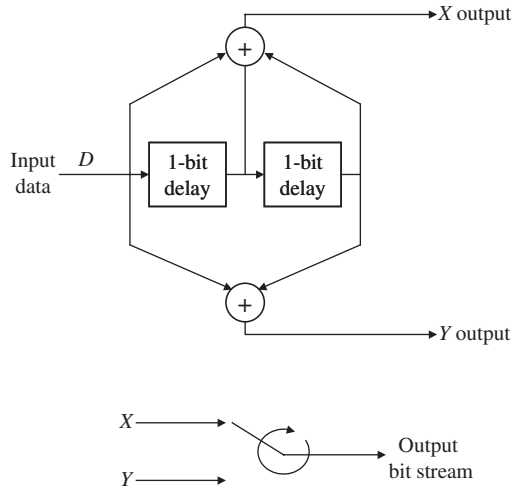


Figure 13.21 Example of 1/2 rate convolutional encoder with constraint length 3.

digital television. The basic structure of a simple convolutional encoder is shown in Figure 13.21. Data arrives at the left of the figure and passes to a shift register, which is clocked once for each arriving bit. For each input bit D , two output bits are generated by calculating the modulo-2 sum of nominated bits in the shift register. Each output bit uses a different combination of these bits. In this example, the output bits are labeled X and Y and are multiplexed into a single output stream $X_1 Y_1 X_2 Y_2 X_3 Y_3 X_4 Y_4 X_5 Y_5 X_6 Y_6 \dots$, whose rate is twice that of the input data stream. Since two output bits are generated for each input bit, this is an example of a 1/2 rate encoder.

Unlike block coders, convolutional coders do not usually have the input bit stream appearing as a subset of the output bits. Even when no errors are introduced into a bit stream, some logic is required in the receiver to recover the original bits. In the case of the simple example above, values of D can be recovered from correctly received X and Y using

$$D_i = X_{i+1} \oplus Y_{i+1}$$

In the more usual case where the decoder must be able to recover from errors, decoding techniques are based on reconstructing possible internal states of the encoder. The Viterbi algorithm provides an optimal method [8].

The major factors that control the performance of a convolutional code are the length of the shift register (usually referred to as the constraint length) and the coding rate. The example shown above can be easily extended to 1/3 rate codes by the addition of another set of taps and exclusive-or function. The use of puncturing, where some bits are deleted before multiplexing at the output of the convolutional encoder, allows lower coding overheads at the cost of some loss of error correction performance. For the example coder, a 2/3 rate code might be produced by deleting every second bit on the X path, leaving a stream: $X_1 Y_1 Y_2 X_3 Y_3 Y_4 X_5 Y_5 Y_6 \dots$

One of the important advantages of convolutional codes is that their performance is less dependent on the location of errors in the bit stream than for a block code. Because there are no block boundaries for convolutional codes, the effects illustrated in Figure 13.20 cannot occur.

The error correcting capability of a convolutional code is not as easily quantifiable as for a block code. Although it is given by

$$t = \left\lfloor \frac{d_f - 1}{2} \right\rfloor$$

where d_f is the minimum free distance of the convolutional code and t is the number of errors that can be corrected within an interval of a few constraint lengths, it is not possible to quantify “a few” without describing the distribution of errors. Generally, “a few” is between three and five.

13.2.5.3. Interleaving

Error correcting codes can be used to detect and correct random bit errors. The codes are effective so long as the number of errors close together remains small. In many types of channel, especially radio channels, however, the channel errors occur in bursts of many errors followed by long periods with almost no errors.

The problem of bursty channel errors can be overcome by interleaving the transmitted data. This is achieved by rearranging the coded data at the transmitter in a predefined pseudorandom order. This means that a burst of errors will be randomized at the receiver when the bits are placed back in their original order.

The simplest form of interleaver is a cyclic interleaver, which operates on fixed-length blocks of data. Each block of input data is read into the register. The output is formed by reading this data out with a different (known) start position. A cyclic interleaver with a start point of zero simply reproduces its input stream.

EXAMPLE 13.2—Cyclic Interleaver

What is the output of a 7-register cyclic interleaver with start position 3?

A 7-register cyclic interleaver with start position 3 and an input sequence 1, 2, 3, 4, ..., 14 produces the output 4, 5, 6, 7, 1, 2, 3, 11, 12, 13, 14, 8, 9, 10. ■

A block interleaver consists of an $M \times N$ block of memory into which arriving bits are scanned in horizontally and scanned out vertically as illustrated in Figures 13.22 and 13.23.

EXAMPLE 13.3—Block Interleaver

What is the output of the 8×3 block interleaver shown in Figure 13.22, in which the data arrives at the interleaver in the order 1 2 3 4 5 6 7 ... 24?

The corresponding scan out is shown in Figure 13.23, showing that data leave the interleaver in the order 1 9 17 2 10 18 3 11 19 4 12 20 5 13 21 ... 8 16 24.

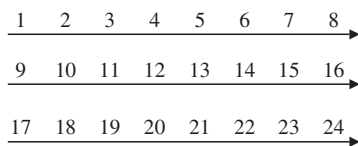


Figure 13.22 Example 8×3 block interleaver — scan in.

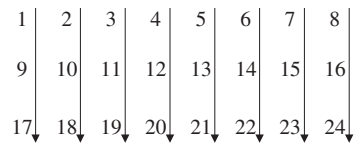


Figure 13.23 Example 8×3 block interleaver — scan out. ■

The convolutional interleaver is more commonly used in digital television systems, and this is illustrated in Figure 13.24.

With $n = 1$ and a 24-element input block with input order 1 2 3 4 5 6 7 8 9 10 ..., the convolutional interleaver illustrated in Figure 13.24 will provide data with output order 1 X X X X X 7 2 X X X X 13 8 3 X X X 19 14 9 4 X X 25 20 15 10 5 X 31 26 21 16 11 6 37 32 27 22 17 12 43 38 33 28 23 18 49 44 39 34 29 24, where X denotes data loaded into the interleaver prior to ‘1.’ The *period* of the convolutional interleaver is the amount of data that is loaded into the shift registers during one cycle of the input data distributor, and it is the product of the number of levels and the size of the registers, denoted by D in Figure 13.24. For the convolutional interleaver shown in Figure 13.24, the period is $6D$.

The performance of a convolutional interleaver is similar to a block interleaver. The primary advantages of the convolutional interleaver are that it causes half the delay and requires only half the memory of a corresponding block interleaver.

The primary purpose of an interleaver is to protect the decoder of a block code from burst errors introduced during transmission. The aim is, therefore, usually to

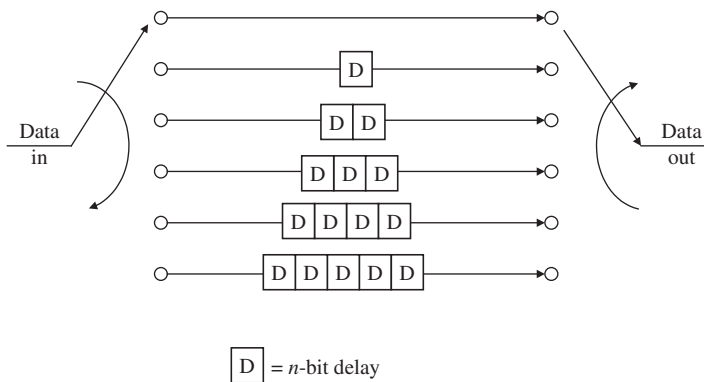


Figure 13.24 Example 6-level convolutional interleaver.

distribute each code word of the block code over several code word transmission times. For example, a 12-level convolutional interleaver with a register size of 17 bytes might be used to protect an (204,188) RS code. Although there is no necessary relationship between the period of the input to the interleaver and the length of the block code word, they often take the same value, which can be used to ensure that the first byte of a code word always passes through the zero-delay path of the interleaver. In this case, the period of the input to the interleaver is 204, which is the same as the code word length. Each 204-byte RS code word is distributed over 12 times its normal transmission interval.

13.2.5.4. Concatenated Codes

Concatenated codes use two levels of coding as illustrated in Figure 13.25. In the encoder, the input data passes through the outer encoder, after which its order is modified by an interleaver before passing through the inner coder. In the decoder, the order of these processes is reversed. The data received from the channel is passed through the inner decoder, which corrects most of the errors introduced during transmission. Where the inner coder fails to correct errors, it is likely to leave bursts of errors at its output. The deinterleaver rearranges the order of the data, spreading these bursts of errors before passing the data to the outer decoder.

Residual errors at the output of the inner decoder tend to be bursty. This would lead to poor performance by the outer decoder, so an interleaver is used to change the order of received data, spreading out bursts of errors to maximize the correcting capability of the outer decoder. A further interleaver may also be used between the inner encoder and the modulator to minimize the likelihood of bursts of errors at the input to the decoder for the inner code.

One of the most popular systems uses a convolutional inner code and an RS outer code. The RS coder is chosen because it can operate on symbols that consist of a number of bits. Like other FEC, it operates best on isolated symbol errors. Because the symbols may consist of a number of bits, the RS coder is quite effective in correcting bursts of bit errors. The interleaver in such systems is likely to be based on the same symbol size as the RS code; there is no value in reordering errors within one symbol of the RS code. Both DVB and ATSC employ such a concatenated coder.

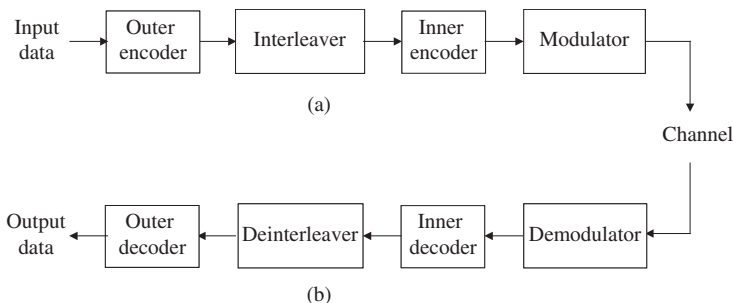


Figure 13.25 Block diagram of (a) a concatenated coder and (b) the corresponding decoder.

13.3. CHANNEL CODING AND MODULATION FOR ATSC

ATSC supports a payload data rate for terrestrial broadcast of 19.28 Mbit/s in one 6 MHz channel [9]. A high data-rate mode is also available that supports a data rate of 38.57 Mbit/s, but this is not intended for use in terrestrial broadcast and is, therefore, not considered further here.

The ATSC coder consists of the following blocks as illustrated in Figure 13.26:

- a data stream randomizer (Section 13.3.3.4);
- a concatenated channel coder (Section 13.3.3), based on:
 - * an RS outer coder,
 - * an outer convolutional interleaver, and
 - * a trellis inner coder (that incorporates an inner block interleaver);
- a framing module in which one segment carries one TS packet, 313 data segments form one data field, and two data fields form one data frame (Section 13.3.2); and
- a modulator, employing 8-VSB modulation (Section 13.3.1).

13.3.1. ATSC 8-VSB Modulation

All data transmitted by an ATSC transmitter is modulated using an 8-VSB modulator at a rate of 10.762 Msym/s, with the eight symbol values denoted by $\{-7, -5, -3, -1, 1, 3, 5, 7\}$. The output of the modulator is passed through a filter to reduce the bandwidth of the transmitted signal to 6 MHz.

The Nyquist criterion tells us that the maximum rate at which symbols can be transmitted over a bandwidth-limited channel and detected by a receiver is equal to twice the channel bandwidth (Section 13.2.1). For a 6 MHz digital television channel, the Nyquist criterion limits the symbol rate to 12 Msym/s, which is higher than the 10.76 Msym/s used by ATSC. As we saw in Section 13.2.1, satisfying the Nyquist criterion does not mean that intersymbol interference will not occur; generally a channel bandwidth at least equal to the symbol rate (that is, twice the Nyquist limit) is required to prevent intersymbol interference. A channel equalizer is, therefore, always required in an ATSC terrestrial receiver. This adaptive equalizer is typically based on the LMS algorithm and employs an FIR filter with approximately 64 taps to equalize received data, with a further 192-tap filter being used in the adaptation process that adjusts the tap values of the 64-tap filter.

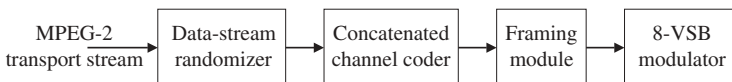


Figure 13.26 Outline structure of ATSC encoder.

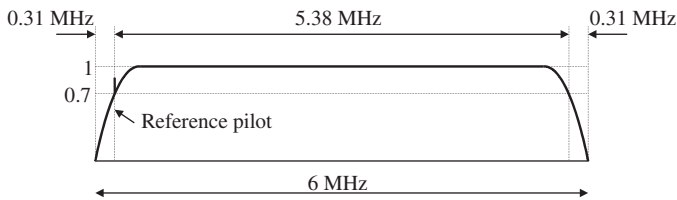


Figure 13.27 Nominal channel occupancy of 8-VSB ATSC transmitted signal. © Advanced Television Standards Committee Inc 2001. A copy of this standard is available at <http://www.atsc.org>.

Transmission of the 10.762 Msym/s, 8-VSB signal in a 6 MHz channel requires filtering at the output of the modulator. The nominal channel occupancy of the filtered signal is shown in Figure 13.27. The half-power bandwidth of the transmitted signal is 5.38 MHz, lying in the center of the 6 MHz channel. A reference pilot signal is transmitted 0.31 MHz from the bottom of the channel, which is the frequency of the suppressed carrier for the VSB signal. A linear-phase, raised-cosine filter is used to produce this band-limited signal.

The primary advantages of 8-VSB are its relatively high bandwidth efficiency and the simplicity created by using only the I component (that is, the value of the symbol does not depend on its Q component). The use of only the I component means that only one analog-to-digital converter is required, and equalization can be carried out using real multiplications and additions.

13.3.2. ATSC Data Framing

The 8-VSB data symbols are grouped into data segments, each consisting of 832 symbols, as illustrated in Figure 13.28. Each data segment is transmitted over 77.3 μ s. The first four symbols of the segment are used for synchronization and always have

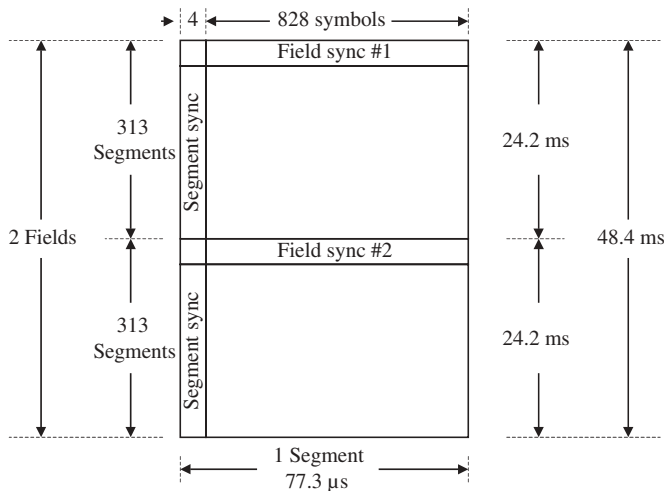


Figure 13.28 ATSC frame structure. © Advanced Television Standards Committee Inc 2001. A copy of this standard is available at <http://www.atsc.org>.

the values $\{5, -5, -5, 5\}$. The remaining 828 symbols (carrying 2484 bits) form the payload of the segment and have exactly the capacity required to carry one MPEG-2 TS packet, complete with all overhead for error correction.

Data segments are grouped into data fields, each consisting of 313 data segments, transmitted over approximately 24 ms. The payload of the first segment of each field contains only a special synchronization sequence, that is, it carries no audio–visual data. This synchronization sequence is used by the decoder to identify the start of a data field and as a training sequence for the channel equalizer. This first segment carries the same 4-symbol synchronization header as all other segments.

A data frame is made up of two data fields and is, therefore, transmitted over approximately 48 ms. The first and second data fields of a data frame are distinguished by the different synchronization sequences carried in their first segments.

13.3.3. ATSC Concatenated Channel Coder

ATSC employs a concatenated coder. The outer coder is an RS coder, which is followed by a convolutional interleaver. The inner coder is a trellis coder that incorporates a stage of inner interleaving.

13.3.3.1. ATSC RS Coder

An $(207,187,8)$ RS code is used to protect each MPEG-2 transport stream packet, allowing up to 10 symbol errors to be corrected in each code word. On entering the RS coder, the length of each transport stream packet is 187 bytes because the sync byte was removed during randomization. The purpose of the outer code is to allow removal of residual errors remaining after decoding of the inner code.

13.3.3.2. ATSC Interleaver

ATSC employs a byte-oriented, 52-level convolutional outer interleaver as shown in Figure 13.29.

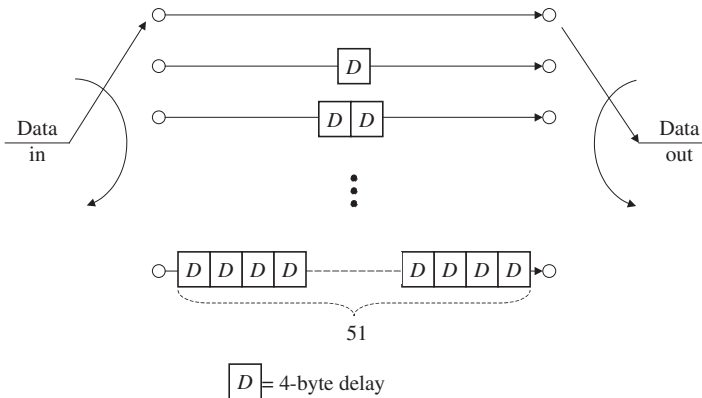


Figure 13.29 ATSC outer interleaver.

This interleaver has a period of 208 bytes, which corresponds to the combined size of the MPEG-2 transport stream sync byte and the outer RS code word. Each input code word is distributed over 52 TS-packet periods, corresponding to approximately 4.0 ms with a data rate of 19.28 Mbit/s or 1/6 of the data field size. The MPEG-2 sync byte is replaced by the 4-symbol data-segment synchronization signal, and always passes through the zero-delay path at the encoder.

In each 207-byte code word, the RS outer coder can correct 10 symbol errors (that is, bytes that contain one or more errors) or 20 erasures. The outer interleaver distributes the 208-byte protected MPEG-2 transport stream packets across 52 TS-packet transmission times in 4-byte groups. The RS coder can, therefore, always correct errors occurring in two of these 4-byte groups within one RS code word, and sometimes correct errors occurring in three or more.

13.3.3.3. ATSC Inner Coder and Interleaver

The ATSC inner coder is a 2/3-rate trellis coder whose input symbols are formed from pairs of output bits from the outer interleaver and whose output consists of 3-bit symbols for the VSB modulator.

At the input to the inner coder, data is formed in bytes. These bytes are then demultiplexed across 12 identical trellis coders (Figure 13.30), whose output is remultiplexed to form the input symbol stream for the 8-VSB modulator. This demultiplexing and remultiplexing is a form of interleaving, whose effect is to spread burst errors on the channel across a number of independent trellis decoders.

Each of the twelve trellis coders has the structure shown in Figure 13.31. Each input byte is broken into four 2-bit symbols, with bits 7, 5, 3, and 1 or the bytes being the more significant bit of their symbol and bits 6, 4, 2, and 0 the less significant bit. The more significant bit (X_2) of each symbol passes through an interference filter precoder. The trellis coder passes the less significant bit (X_1) of each symbol through a 1/2 rate convolutional encoder but leaves the more significant bit uncoded. The outputs of the precoder (Z_2) and convolutional encoder (Z_1 and Z_0) then pass to the symbol mapper that converts the 3-bit symbol inputs into one of the eight levels at its output. This mapping, in which the distance Z_2 is more significant than Z_1 and Z_0 , explains why it is reasonable that redundancy is added to X_1 but not to X_2 .

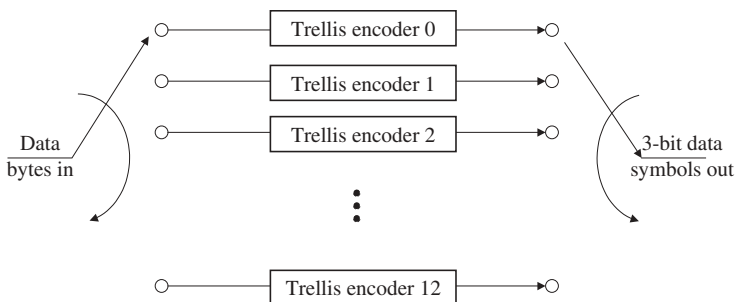


Figure 13.30 ATSC inner interleaver.

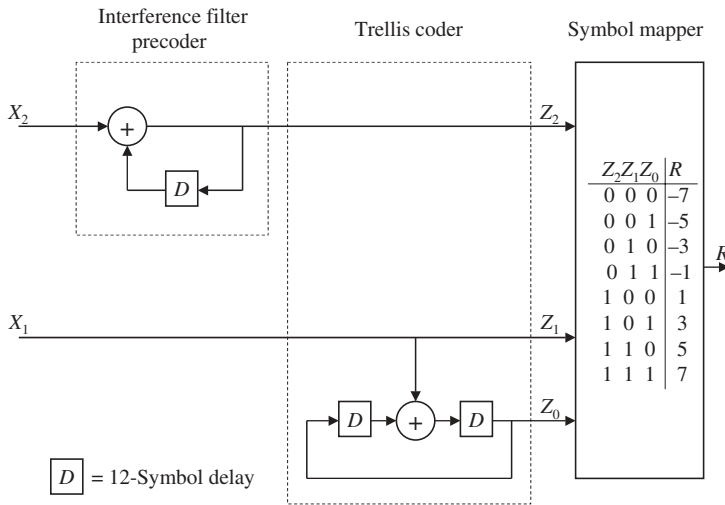


Figure 13.31 VSB trellis coder with interference filter precoder and symbol mapper. © Advanced Television Standards Committee Inc 2001. A copy of this standard is available at <http://www.atsc.org>.

13.3.3.4. ATSC Randomization

Data stream randomization in ATSC is applied only to the 187-byte payloads of MPEG-2 TS packets. The sync byte of MPEG-2 TS packets is discarded and replaced by the data segment sync signal. The randomization is carried out by taking the exclusive-or of the payload with the output of a pseudorandom binary sequence generator, which is initialized at the beginning of every data field. This initialization, combined with the unique data field sync signal that occupies the first data segment of the data field, makes random access possible at the beginning of any data field, in other words, random access is possible once in every 24 ms.

The pseudorandom sequence generator is a 16-bit linear shift register with eight feedback taps and eight outputs as shown in Figure 13.32. Each element in the shift register is labeled with its index. At each clock cycle, one output byte $[D_7 \dots D_0]$ is produced. This output byte is bit-wise exclusive-ored with one byte of the MPEG-2 TS packet. The shift register is initialized to the value 0xF180 at the beginning of every data field (that is, registers 16, 15, 14, 13, 9, and 8 are loaded with 1, the other registers with 0).

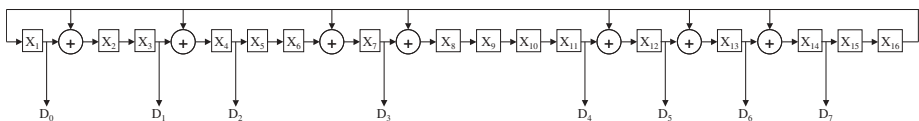


Figure 13.32 Detailed structure of ATSC PN generator. © Advanced Television Standards Committee Inc 2001. A copy of this standard is available at <http://www.atsc.org>.

13.3.4. ATSC Channel Capacity

The capacity of the ATSC channel can be derived as follows. The raw symbol rate on the channel is 10.762 Msym/s. Each symbol carries 3 bits of data. The trellis coder is a rate $2/3$ coder, so that two out of every three bits transmitted are information (the other is overhead). This rate is reduced by the overheads for transmission of field sync segments (312/313) and the outer RS code (188/208). The useful transmission capacity is, therefore,

$$U = 10.762 \times 3 \times \frac{2}{3} \times \frac{312}{313} \times \frac{188}{208} = 19.392 \text{ Mbit/s}$$

This value for U includes the MPEG-2 transport stream sync bytes, which are mapped to the segment sync headers before transmission. This method of calculation differs from that used in the ATSC standard, but it provides a value that is directly useable in assessing the channel capacity in terms relevant to the MPEG-2 transport stream. ATSC documents often quote a useful data rate of 19.29 Mbit/s, which excludes the MPEG-2 transport stream sync bytes.

13.4. CHANNEL CODING AND MODULATION FOR DVB

The DVB channel coder takes an MPEG-2 transport stream as input and passes it through the following processes prior to modulation [10] as illustrated in Figure 13.33:

- transport adaptation and randomization,
- outer coding using Reed–Solomon code,
- outer interleaving,
- inner coding using a punctured convolutional code, and
- inner interleaving.

The output of the channel coder is passed to the modulator for modulation using QPSK, 16-QAM, or 64-QAM and orthogonal frequency-division multiplexing. Each of these processes is described in the following sections.

13.4.1. DVB Modulation

Having completed the processes of outer coding and interleaving and inner coding and interleaving, a DVB encoder proceeds to modulation. Data is modulated using QPSK, 16-QAM, or 64-QAM. The first step of the modulation process is to group

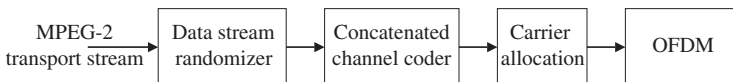


Figure 13.33 Outline structure of DVB channel coding and modulation.

the data bits to form complex-valued modulation symbols, denoted by z , based on the constellation diagrams in Section 13.2.1. For QPSK the constellation diagram in Figure 13.5 is used. Each QPSK symbol is made of 2 bits. The symbol value obtained from Figure 13.5 is then normalized so that the average symbol energy is 1. For QPSK, this requires dividing the symbol value by $\sqrt{2}$, giving a normalized symbol value c . For example, the symbol value z for the bits 10 is $z = -1 + j$, where j denotes the square root of -1 , and $c = -0.7071 + 0.7071j$.

Figure 13.6 shows the corresponding constellation diagram for 16-QAM, for which each symbol consists of 4 bits. Normalization for 16-QAM requires dividing each symbol value by $\sqrt{10}$. Figure 13.7 shows the constellation diagram for the 6-bit symbols used for 64-QAM, for which normalization requires division of the symbol value by $\sqrt{42}$.

The normalized complex symbols are then converted to a sequence of sample values using orthogonal frequency-division multiplexing (OFDM), during which pilot signals and signaling information are incorporated into the signal.

13.4.1.1. DVB Orthogonal Frequency-Division Multiplexing

The basic idea of orthogonal frequency-division multiplexing (OFDM) is to split the data to be carried on a channel among several subchannels, and to frequency-division multiplex these subchannels within the bandwidth of the original channel. A very simple implementation, using four subchannels, is illustrated in Figure 13.34.

Figure 13.34(a) shows a block diagram of the four subchannel OFDM modulator. Figure 13.34(b) shows the approximate power spectrum for a conventional signal, whereas Figure 13.34(c) shows the power spectrum of the OFDM signal. The OFDM signal provides a more even distribution of power across the bandwidth of the channel and has sharper drop off in power at the edge of the channel, which simplifies the design of transmitter output filters.

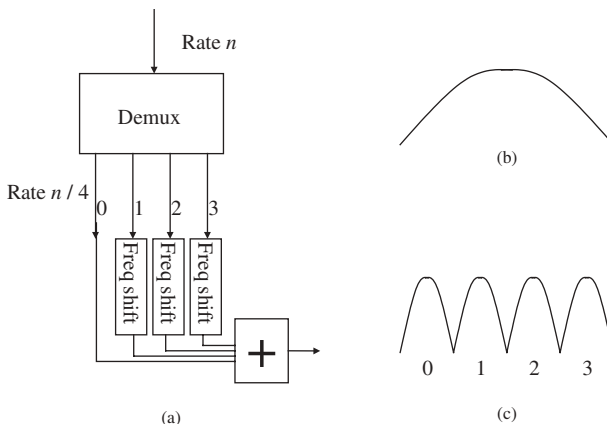


Figure 13.34 Simple OFDM with four carriers.

The major requirements for an effective implementation of OFDM are that

- the subchannels must be synchronized so that remultiplexing is possible at the decoder to reconstruct exactly the original bit stream, and
- the modulation waveforms for the individual subchannels should be orthogonal to provide high spectral efficiency.

Under these circumstances, the principal advantages of OFDM are

- *spectral efficiency*—signal is more evenly spread over channel than for a single data stream;
- *immunity to fading and interference*—fading or interference associated with one modulated carrier may not affect others; when data is protected by FEC, this provides a powerful means for overcoming narrowband fading and interference; and
- *interference with other signals*—minimization of signal level outside channel is easily achieved without complicated filtering.

The DVB OFDM system has two modes of operation: the 2k mode and the 8k mode. In the 2k mode, there are 2048 notional carriers spread across the whole channel bandwidth, with constant intercarrier spacing. Only a subset of the carriers (1705) at the center of the channel is actually used for transmission. The remaining notional carriers are not generated and, therefore, help with suppression of out-of-channel emissions. The signals transmitted on the 2048 carriers are each orthogonal to every other signal (i.e., their cross correlation is zero), enabling high spectral efficiency.

For the 8k mode, there are 8192 notional carriers spread evenly across the channel bandwidth. Of these carriers, the 6817 lying in the middle of the channel bandwidth are used for transmission. Once again the signals transmitted on the carriers are orthogonal, enabling high-spectral efficiency.

Each carrier may be modulated by QPSK, 16-QAM, or 64-QAM. The group of symbols transmitted simultaneously on all carriers is called an OFDM symbol. OFDM symbols are grouped together to form OFDM frames, each of which contains 68 OFDM symbols. Frames are grouped to form super frames, each of which contains four frames.

More formally, the output s of the OFDM modulator is given by

$$s(t') = \text{Re} \left\{ e^{j2\pi f_c t'} \sum_{k=k_{\min}}^{k=k_{\max}} c_{m,l-1,k} e^{j2\pi k'(t'-\Delta)/T_U} \right\}$$

where t' is the time relative to the beginning of the symbol transmission period; T is the elementary period, which is the reciprocal of the clock period; k denotes the carrier number; l denotes the OFDM symbol number; m denotes the transmission frame number; K is the number of transmitted carriers (1705 for the 2k mode and 6817 for the 8k mode); T_U is the inverse of the spacing between OFDM carriers; ($T_U = 2048 \times T$ for the 2k mode and $T_U = 8192 \times T$ for the 8k mode); Δ is the

duration of the guard interval; $T_S = \Delta + T_U$ is the symbol duration; f_c is the center frequency of the RF signal; $K_{\min}$ is the index of the lowest frequency carrier (0 for both 2k and 8k modes); $K_{\max}$ is the index of the highest frequency carrier (1704 for 2k mode, 6816 for 8k mode); $k' = k - (K_{\max} - K_{\min})/2$ is carrier index relative to the channel center frequency f_c ; $c_{m,l-l,k}$ is the normalized complex symbol for carrier k of the data symbol number l in frame m .

For a channel bandwidth of 7 MHz, the clock frequency is 8 MHz and the elementary period $T = 1/8 \mu\text{s}$. DVB also supports operation using 6 MHz and 8 MHz channel bandwidths. The only changes required to encoder and decoder operation are in the clock frequency (and therefore the elementary period T). For a 6 MHz channel, the clock frequency is 48/7 MHz and the elementary period $T = 7/48 \mu\text{s}$. For an 8 MHz channel, the clock frequency is 64/7 MHz and the elementary period $T = 7/64 \mu\text{s}$. A 6 MHz channel, therefore, operates at 6/7 the symbol rate of a 7 MHz channel and has 6/7 the useful bit rate. An 8 MHz channel operates at 8/7 the symbol rate of a 7 MHz channel and has 8/7 the useful bit rate.

In converting from the OFDM symbol to a time sequence s to be transmitted, the sampling rate must be at least twice the bandwidth of the symbol to ensure that aliasing does not occur; in other words, the sampling rate for s must be greater than or equal to $2/T$. One common implementation of this conversion is by means of an inverse Fourier transform. The choice of the number of notional carriers as a power of two in both the 2k and 8k modes facilitates this transform, making possible the use of the inverse fast Fourier transform (IFFT). The principle advantage of the IFFT over other implementations of the transform is that its computational complexity is of the order of $(n \log n)$, rather than of the order of (n^2) , where n is the number of carriers.

In the 2k mode, DVB uses 1705 carriers in each OFDM symbol, of which 193 are used for pilots and signaling (described in Sections 13.4.1.4 and 13.4.1.5 below), leaving 1512 useful carriers. In the 8k mode, there are 6817 carriers in each OFDM symbol, with 769 used for pilots and signaling, leaving 6048 useful carriers.

13.4.1.2. DVB Guard Interval

The choice of the intercarrier spacing (which is effectively the channel bandwidth for the stream of signals transmitted on one carrier) to be the reciprocal of the OFDM symbol duration T_U means that essentially all the intersymbol interference in a DVB system is caused by the channel. The purpose of the guard interval is to reduce the impact of intersymbol interference by increasing the spacing between symbols and allowing time for the impulse response of the channel to decay between the point where the encoder changes symbol value and the point where the decoder uses the received signal in demodulation. The DVB guard interval is created by making a copy of that portion of symbol to be transmitted last and transmitting this immediately in front of the symbol as shown in Figure 13.35. The size of the DVB guard interval determines what fraction of the symbol is copied in this way. In DVB, the guard interval may be chosen to be 1/4, 1/8, 1/16, or 1/32 of the OFDM symbol period, and it would usually be chosen to ensure that the impulse response of the channel has decayed before the beginning of the symbol to be decoded. This is especially

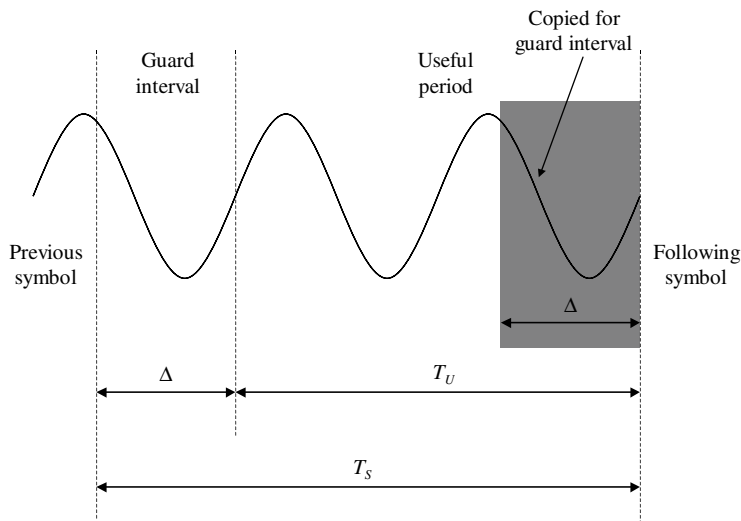


Figure 13.35 Construction of DVB guard interval.

important for channels in which the signal travels between transmitter and receiver over multiple, simultaneous paths (i.e., multipath channels). Without this guard interval, multipath propagation leads to significant intersymbol interference. Because the OFDM symbol period is longer for the 8k mode than that for the 2k mode, the 8k mode is more suitable for channels with longer impulse responses than that for the 2k mode. One of the major advantages of this use of the guard interval is that the requirement for channel equalization is reduced to one complex multiplication for each OFDM carrier if the decoder in signal processing is based on a Fourier transform (i.e., reversing the inverse Fourier transform at the encoder).

The choice of guard interval length directly affects the useful channel capacity available. Table 13.2 shows how the number of OFDM symbols per second varies

Table 13.2 Impact of guard interval on number of OFDM symbols per second.

Mode	Δ/T_U	OFDM symbols per second	Loss of capacity due to guard interval (%)
2k	1/4	3125	20
	1/8	3472	11
	1/16	3677	6
	1/32	3788	3
	0 ^a	3906	0
8k	1/4	781	20
	1/8	868	11
	1/16	919	6
	1/32	947	3
	0 ^a	977	0

^aNot supported by DVB. The number of OFDM symbols per second is $1/T_U$ in this case.

with the mode (2k or 8k) and size of guard interval (Δ/T_U) for a 7 MHz channel. Changing between modes does not affect the relative loss of capacity for fixed Δ/T_U . It is likely, however, that a smaller value of Δ/T_U would be used for a given channel in the 8k mode than in the 2k mode, so that the length of the guard interval Δ is the same. This means that for a fixed transmission channel, less loss of capacity is likely to be required for guard intervals in the 8k mode than in the 2k mode. Alternatively, the 8k mode allows operation with channels whose impulse response takes longer to decay than is possible in the 2k mode.

13.4.1.3. Single-Frequency Network

In conventional television broadcast, black spots of extremely poor reception occur due to terrain. Additional, local repeaters are sometimes used to provide an acceptable signal quality in such areas. Because of their potential to cause interference with other signals, it is usual that such repeaters are allocated a separate channel from the main transmitter. Sometimes, it is even necessary to allocate a separate channel to each repeater in an area. This is a very inefficient use of the limited bandwidth available for terrestrial-broadcast television. The use of OFDM in combination with channel coding (known as *coded OFDM*) in DVB offers the possibility of operating a large number of repeaters on one channel. Interference will occur on a small proportion of the carriers, causing fading and very high bit error rates. The combination of inner and outer coding, spread across a large number of carriers by interleaving, overcomes this interference, leading to a high-quality bit stream at the input to the MPEG-2 systems decoder. This is known as a single-frequency network.

A single-frequency network is illustrated in Figure 13.36. Two transmitters (“A” and “B”) are used. Clearly, the use of two transmitters in close proximity, operating on the same frequency, can lead to interference between their signals. The difference between the propagation delays from the two transmitters to the receiver is the major potential cause of degradation in the received signal.

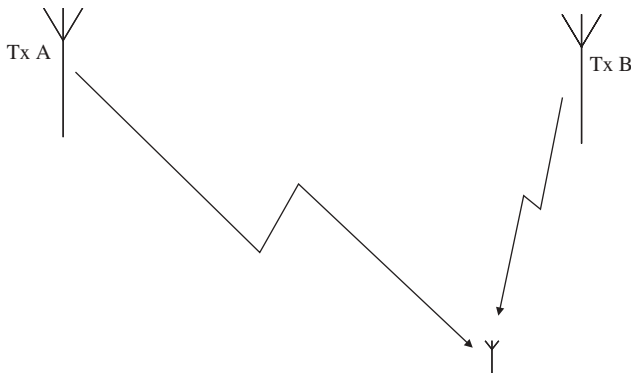


Figure 13.36 A simple single-frequency network.

The first impact of the difference in propagation delay is that it can cause inter-symbol interference. So long as the difference in propagation delays from the two transmitters is less than the DVB guard interval, the contribution of the single-frequency network to intersymbol interference should be negligible.

The second effect of receiving two signals from different sources is that the received power depends on the relative phase of the two signals, which changes with frequency. If the phase difference between the two signals is one half cycle, destructive interference causes the received power to be zero. In analyzing the impact of this interference, we assume the worst case, that is, that the signal strengths from A and B are the same. In this case, nulls occur at frequencies where the difference in path lengths between the transmitters and the receiver takes particular values satisfying

$$d_{null} = \frac{\lambda(2n+1)}{2}$$

for integer values of n . The frequency spacing between these nulls is equal to the reciprocal of the difference in propagation delay from A to the receiver and B to the receiver.

If the signals received from transmitters A and B have the same electric field strength, we can write

$$E_A = \sin(2\pi ft + \phi/2)$$

and

$$E_B = \sin(2\pi ft - \phi/2)$$

where ϕ is the phase shift caused by the difference in propagation delays between A and the receiver and B and the receiver. The total electric field strength of the received signal is therefore

$$E_{A+B} = 2\cos(\phi/2) \sin(2\pi ft)$$

In other words, the amplitude of the received signal is $2\cos(\phi/2)$. The implications of this calculation are that

- the peak received power is four times higher than that for one transmitter,
- the received power is higher than that would be received from a single transmitter for 2/3 of the OFDM carriers, and
- the received power is higher than half that of what would be received from a single transmitter for 77 % of the OFDM carriers.

In a DVB system, the OFDM carriers for which the received power is small tend to be clustered in groups. It is for this reason that DVB does not assign symbols to OFDM carriers simply in order of increasing frequency, but assigns symbols to carriers in a pseudorandom order.

Table 13.3 Relationship between received power and signal cancellation in a single-frequency network.

Received power (dB)		Peak power (dB)	Carriers with power ≥ 0 dB	Carriers with ≥ -3 dB	Carriers with ≥ -6 dB
A	B				
0	0	6.0	66 %	77 %	84 %
0	-3	4.6	61 %	75 %	85 %
0	-6	3.5	58 %	77 %	100 %
0	-12	1.9	53 %	100 %	100 %

EXAMPLE 13.4—Single-Frequency Network

Calculate the difference in propagation delay experienced by two signals in a single-frequency network where one transmitter is 3 km closer to the receiver than the other. Calculate also the spacing between frequencies for which the received signal strength is zero if the received power from each of the two transmitters is the same.

A difference in path length of 3 km is equivalent to a 10 μ s difference in propagation delay. The frequency spacing between nulls in the received power is 100 kHz. ■

In the more general case where the strengths of the signals received from the two transmitters differ, the peak power is less than that indicated by the above analysis; but the received power never drops to zero. The proportion of the OFDM carriers for which the power is less than half the stronger of the received powers is also reduced. This is illustrated in Table 13.3 and Figure 13.37.

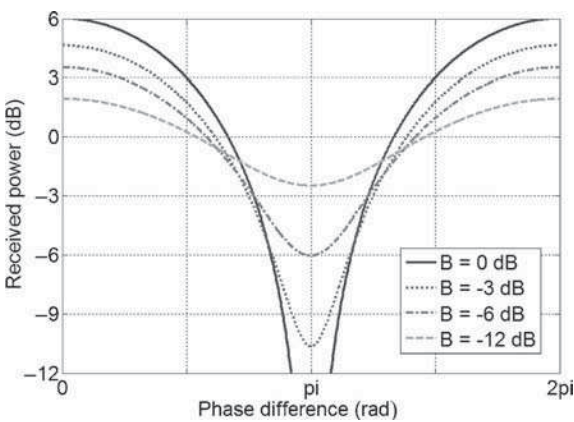


Figure 13.37 Total received power versus phase difference for 0 dB received from A and various powers from B.

Table 13.4 Carrier indices for TPS.

2k Mode	8k Mode
34 50 209 346 413 569	34 50 209 346 413 569 595 688 790 901 1073 1219 1262 1286
595 688 790 901 1073	1469 1594 1687 1738 1754 1913 2050 2117 2273 2299 2392
1219 1262 1286 1469	2494 2605 2777 2923 2966 2990 3173 3298 3391 3442 3458
1594 1687	3617 3754 3821 3977 4003 4096 4198 4309 4481 4627 4670
	4694 4877 5002 5095 5146 5162 5321 5458 5525 5681 5707
	5800 5902 6013 6185 6331 6374 6398 6581 6706 6799

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13.4.1.4. DVB Transmission Parameter Signaling

A DVB decoder is required to be able to identify the parameters of a received signal and begin decoding it without user intervention. This means that the decoder must identify features, such as the type of modulation, the inner coding rate, and guard interval, which cannot be easily inferred from the encoded data itself. *Transmission parameter signaling (TPS)* is used to provide this type of information to the decoder in an easily accessible way.

TPS data is carried on a number of dedicated OFDM carriers, 17 for the 2k mode and 68 for the 8k mode, each of which carries an identical data stream. The indices of these carriers are shown in Table 13.4. These carriers are never used for any other purpose. All TPS data is modulated using differential BPSK (DBPSK), so that a decoder does not need to identify the modulation type before decoding this data.

Each OFDM symbol carries one TPS bit. Sixty eight consecutive OFDM symbols form an OFDM frame. The 68 bits of TPS data is carried by each OFDM frame form one TPS block. Each TPS block contains one initialization bit, followed by 16 synchronization bits, 37 information bits, and 14 redundancy bits for error protection. The contents of each field are described in Table 13.5.

13.4.1.5. DVB Reference Signals

Of the 1705 transmitted carriers in the 2k mode, 1512 are available for carrying data and the remaining 193 are used to carry reference signals. (Similarly, in the 8k mode, there are 6048 useful carriers and 769 are used for reference signals.) In addition to TPS, DVB uses two other types of reference signals: continual pilots and scattered pilots. Both continual and scattered pilots transmit a pseudorandom sequence, which is modulated using BPSK.

Scattered pilots are always transmitted on the lowest and highest frequency carriers. They are also transmitted on every twelfth carrier, starting at $K_{\min}$ for symbols 0, 4, 8, ..., 64 in a frame; $K_{\min}+3$ for symbols 1, 5, 9, ..., 65; $K_{\min}+6$ for symbols 2, 6, 10, ..., 66; and $K_{\min}+9$ for symbols 3, 7, 11, ..., 67. Continual pilots are always transmitted on the same carriers, which are listed in Table 13.6.

Table 13.5 Usage of fields in each TPS block.

Field purpose	Length	Description
Initialization bit	1	Initialization bit for DBPSK.
Synchronization bits	16	Alternating 0011 0101 1110 1110 and 1100 1010 0001 0001.
Length indicator	6	010 111 if cell identifier not supported, 011 111 otherwise.
Frame number	2	Frame number in superframe.
Constellation	2	Identifies QPSK, 16-QAM, or 64-QAM as the modulation used for data.
Hierarchy information	3	Selects hierarchical/nonhierarchical mode.
Code rate, HP stream	3	Selects inner code rate from 1/2, 2/3, 3/4, 5/6, and 7/8 for nonhierarchical or high-priority stream of hierarchical mode.
Code rate, LP stream	3	Selects inner code rate from 1/2, 2/3, 3/4, 5/6, and 7/8 low-priority stream of hierarchical mode.
Guard interval	2	Identifies guard interval from 1/32, 1/16, 1/8, and 1/4.
Transmission mode	2	Selects between 2k mode and 8k mode. This field must be guessed first by the receiver, but transmission in TPS can be used to confirm this guess.
Cell identifier	8	Used to identify the cell from which the signal comes.
Reserved for future use	6	Set all to zero.
Error Protection	14	BCH code protecting the TPS block.

Table 13.6 Carrier indices for continual pilots.

2k Mode	8k Mode
0 48 54 87 141 156 192 201	0 48 54 87 141 156 192 201 255 279 282 333 432 450 483 525
255 279 282 333 432 450	531 618 636 714 759 765 780 804 873 888 918 939 942 969
483 525 531 618 636 714	984 1050 1101 1107 1110 1137 1140 1146 1206 1269 1323
759 765 780 804 873	1377 1491 1683 1704 1752 1758 1791 1845 1860 1896 1905
888 918 939 942 969	1959 1983 1986 2037 2136 2154 2187 2229 2235 2322
984 1050 1101 1107 1110	2340 2418 2463 2469 2484 2508 2577 2592 2622 2643
1137 1140 1146 1206	2646 2673 2688 2754 2805 2811 2814 2841 2844 2850
1269 1323 1377 1491	2910 2973 3027 3081 3195 3387 3408 3456 3462 3495
1683 1704	3549 3564 3600 3609 3663 3687 3690 3741 3840 3858
	3891 3933 3939 4026 4044 4122 4167 4173 4188 4212 4281
	4296 4326 4347 4350 4377 4392 4458 4509 4515 4518 4545
	4548 4554 4614 4677 4731 4785 4899 5091 5112 5160 5166
	5199 5253 5268 5304 5313 5367 5391 5394 5445 5544 5562
	5595 5637 5643 5730 5748 5826 5871 5877 5892 5916 5985
	6000 6030 6051 6054 6081 6096 6162 6213 6219 6222
	6249 6252 6258 6318 6381 6435 6489 6603 6795 6816

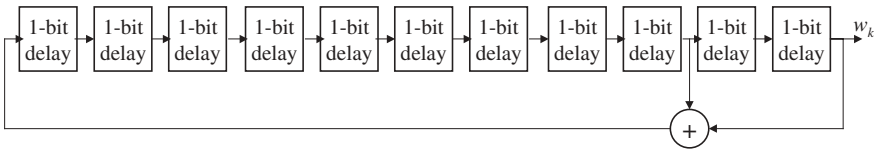


Figure 13.38 Pseudo random sequence generator for pilot signals.

The outline structure of the pseudorandom sequence carried by both continual and scattered pilots is shown in Figure 13.38. The generator is reset at the beginning of every OFDM symbol by setting all elements of the shift register to 1. One bit w_k is generated for each carrier of the OFDM symbol, regardless of whether this bit carries data or a pilot signal; that is, if carrier k carries a pilot signal, the value of the pilot is w_k .

13.4.1.6. DVB Spectrum Characteristics

DVB supports channel bandwidths 6, 7, and 8 MHz. One important advantage of the use of OFDM is that changing from one channel bandwidth to another requires only scaling the elementary period T in proportion to the new channel bandwidth required.

Figure 13.39 shows the power spectrum of a DVB transmission signal for an 8 MHz channel in the 2k mode with a guard interval $\Delta = T_U/4$. Frequency is shown relative to the center frequency f_c of the channel. The power spectral density is simply the sum of the power spectral densities of the individual modulated carriers. In this mode of operation, the power spectral density of the transmitted signal is attenuated by more than 30 dB outside the allocated channel. The frequency of the highest frequency carrier is approximately 3.83 MHz above the channel center frequency, and the frequency of the lowest frequency carrier is approximately 3.83 MHz below the center frequency.

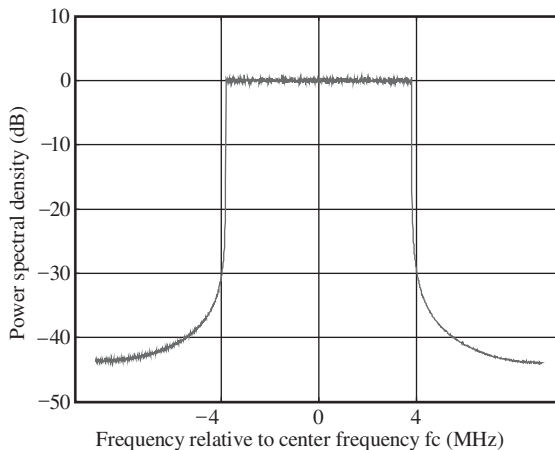


Figure 13.39 DVB transmission signal spectrum for 8MHz channel operating in 2k mode with guard interval $\Delta = T_U/4$.

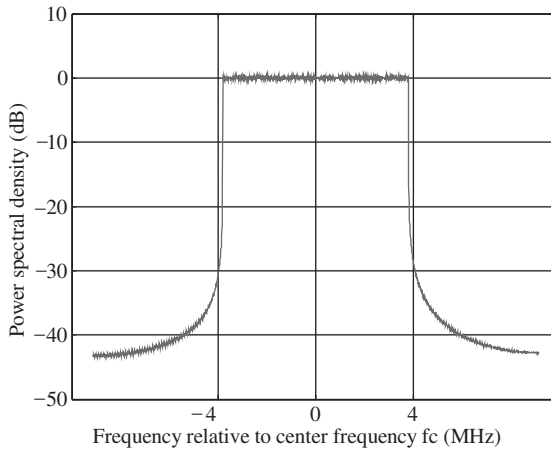


Figure 13.40 DVB transmission signal spectrum for 8 MHz channel operating in 2k mode with guard interval $\Delta=T_U/32$.

Alteration of the guard interval to $\Delta=T_U/32$ produces the power spectral density shown in Figure 13.40. There are no significant differences between this spectrum and that shown previously for $\Delta=T_U/4$. This is because the change in guard interval makes no difference to the center frequencies of OFDM carriers. A change in the line width around individual carriers does occur, because a change in guard interval changes the number of OFDM symbols per second (Table 13.2). The line width around individual carriers, therefore, may vary by up to 20%, but the overall impact on the bandwidth of the channel is negligible.

Changing from the 2k mode to the 8k mode increases the attenuation at the edge of the channel's allocated bandwidth by approximately 6–36 dB as shown in Figure 13.41.

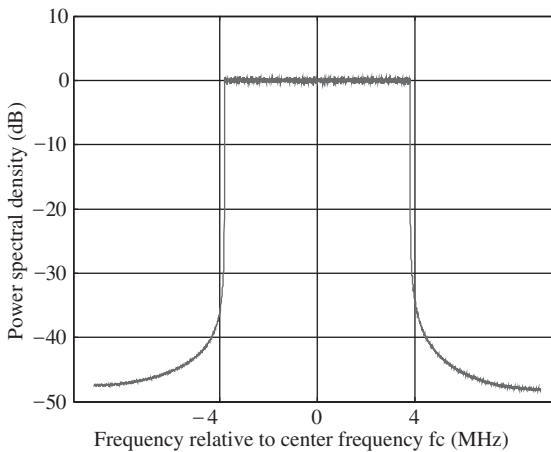


Figure 13.41 DVB transmission signal spectrum for 8 MHz channel operating in 8k mode with guard interval $\Delta=T_U/4$.

This occurs because the OFDM symbol rate in the 8k mode is one quarter of that in the 2k mode, causing the bandwidth of individual modulated carriers in the 8k mode to be one quarter of that in the 2k mode leading to a reduction in their contribution to out-of-band power.

Other properties of the DVB transmitted signal, such as modulation type and inner code rate, do not significantly affect the transmitted power spectral density.

13.4.2. DVB Channel Coding

The DVB channel coder employs data randomization, an RS outer coder, an outer convolutional interleaver, an inner convolutional coder, and an inner block interleaver.

13.4.2.1. DVB Randomization

Transport-stream randomization in DVB is performed using a pseudorandom sequence generated by a 15-stage shift register shown in Figure 13.42. The pseudorandom sequence generator is initialized with the value 100101010000000 and used to generate 1503 bytes of data. These bytes are bit-wise exclusive ored with eight transport-stream packets, starting immediately after the sync byte of the first transport-stream packet. Sync bytes of transport-stream packets are not randomized, but 1 byte of the sequence is skipped for the sync byte of all subsequent transport-stream packets. The first transport-stream packet in the group of eight is identified in the transmitted bit stream by having its sync byte inverted (that is, 0x47 is transmitted as 0xB8).

13.4.2.2. DVB Outer Coding

DVB's outer coder uses an RS (204, 188, 8) code, with the information part of each code word representing one MPEG-2 transport stream packet. This code can correct up to eight symbol errors in each received code word. The construction of the code word is illustrated in Figure 13.43.

Each byte of the data to be protected is treated as one symbol. The first step is to prefix the 188-symbol transport stream packet with 51 symbols containing all zeros, giving a 239-symbol block. The parity symbols for this block are then calculated as

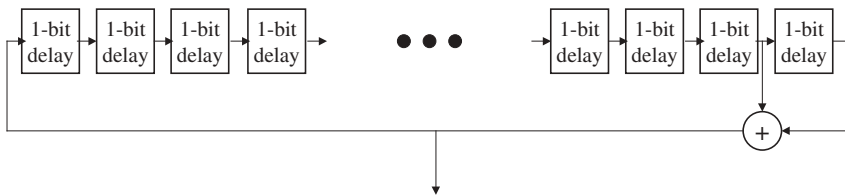


Figure 13.42 Structure of the pseudorandom sequence generator for DVB transport randomization.

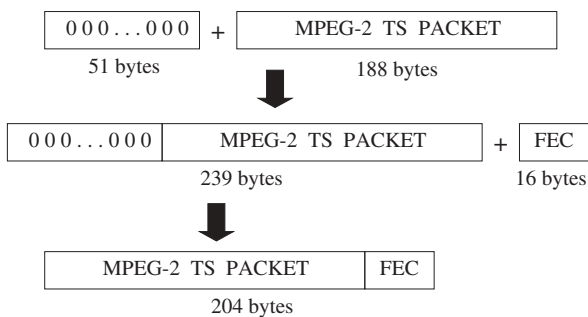


Figure 13.43 DVB outer coding.

for an RS (255,239) code, generating a 16-symbol parity block. The 51 zero symbols are then discarded, leaving a 204-symbol code word, made up of the 188-byte transport stream packet and 16 parity bytes.

13.4.2.3. DVB Outer Interleaving

The DVB outer interleaver is a 12-level, byte-oriented convolutional interleaver as shown in Figure 13.44. Each delay element D corresponds to a delay of 17 bytes, leading to a period of 204 bytes. The sync byte of the first MPEG-2 TS packet is always routed through the zero-delay path, causing the sync bytes of all future TS packets to also be routed through the zero-delay path (Table 13.7).

This interleaver is byte oriented because the RS code it precedes in the decoder treats each byte of data as a single symbol. Where burst errors occur, it is therefore desirable to keep these errors within the symbols in which they occur, rather than spread them to other symbols and thereby increase the symbol error rate.

In each 204-byte code word, the RS outer code can correct eight symbol errors (i.e., bytes that contain one or more errors) or 16 erasures. The outer interleaver distributes the 204-byte RS code words across 12 TS-packet transmission times in 17-byte groups.

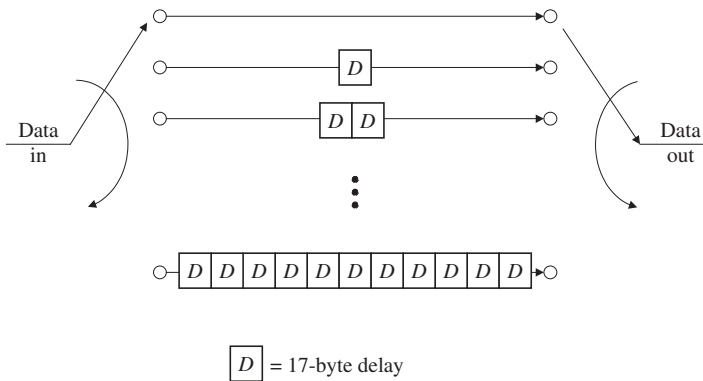


Figure 13.44 DVB outer interleaver.

Table 13.7 Temporal spread of an interleaved TS packet in a DVB transmission signal.

Useful bit rate (Mbit/s)	Interleaving spread (ms)
5	3.6
10	1.8
15	1.2
20	0.9
25	0.7
30	0.6

13.4.2.4. DVB Inner Coding

The DVB inner coder is based on the 1/2 rate convolutional coder with 64 states (i.e., a constraint length of seven) as shown in Figure 13.45. The data arrives at the left side of the figure and passes through the 6-bit shift register. For each input bit, the contents of the shift register are shifted one position to the right side and two output bits, *X* and *Y*, are generated. These two output bits are then multiplexed to provide two bits for the single-coded output bit stream. This particular code offers the best performance available from a 1/2 rate, constrain-length seven convolutional code [11]. This code has a free distance d_f equal to 10, meaning that it can correct bursts of up to four errors, following the discussion of Section 13.2.5.2.

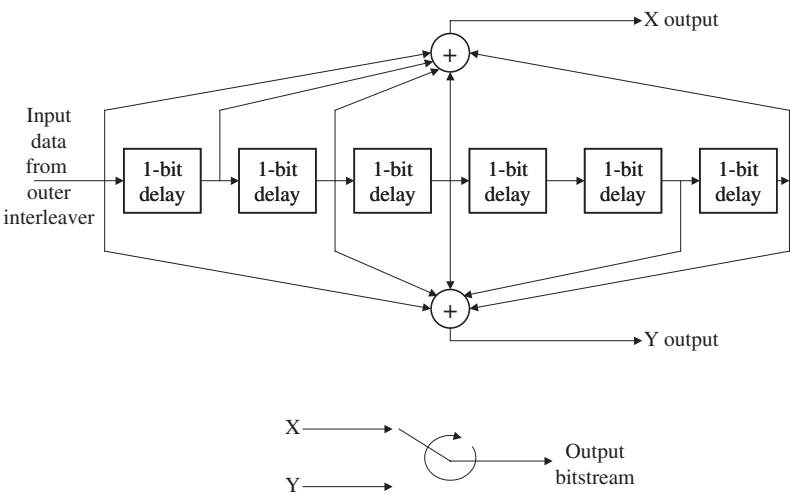


Figure 13.45 The 1/2 rate convolutional coder used for DVB’s inner coder. © European Telecommunications Standards Institute 1997. © European Broadcasting Union 2004. Further use, modification, redistribution is strictly prohibited. ETSI standards are available from <http://pda.etsi.org/pda/> and http://www.etsi.org/services_products/freestandard/home.htm.

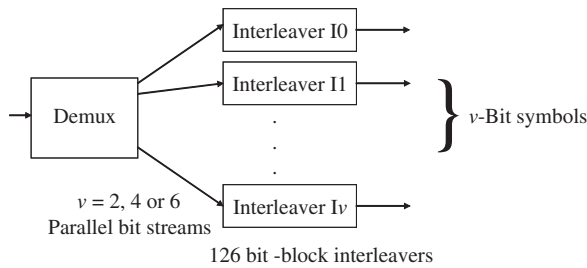


Figure 13.46 DVB inner interleaver.

For operation at lower coding rates, puncturing is used, allowing coding rates of $2/3$, $3/4$, $5/6$, and $7/8$ to be achieved. Puncturing is commonly used with convolutional codes to allow a variety of different levels of error protection to be achieved, without requiring more than one decoder to be implemented. Changing the error protection overhead with block codes, in contrast, requires the use of a different code and decoder for each level of protection.

13.4.2.5. DVB Inner Interleaving

The purpose of inner interleaving is to randomize the order of errors being passed into the inner (convolutional) decoder. The first stage of inner interleaving is to break the bit stream into v -bit symbols for modulation. For QPSK, $v = 2$; for 16-QAM, $v = 4$; and for 64-QAM, $v = 6$. Each bit of the v -bit symbols is passed through a bit-wise interleaver, after which the symbols are reconstructed and assigned to OFDM carriers by the symbol interleaver.

The outline structure of the inner interleaver is shown in Figure 13.46. Up to six different block interleavers are used depending on the modulation (QPSK, 16-QAM, or 64-QAM). Each of the bit-wise interleavers $I_0 \dots I_5$ operates on a 126-bit block and performs a cyclic shift of this block. The starting points for the cyclic shift are shown in Table 13.8.

The symbol interleaver is responsible for assigning interleaved data symbols to OFDM carriers. Each OFDM symbol carries 12 126-symbol blocks in the 2k mode and 48 126-symbol blocks in the 8k mode.

Table 13.8 Start points for 126-bit cyclic interleavers.

Interleaver	Start point
I0	0
I1	63
I2	105
I3	42
I4	21
I5	84

13.4.3. DVB Channel Capacity

The useful transmission capacity of a DVB channel depends on the elementary period, the modulation (QPSK, 16-QAM, or 64-QAM), the length of the guard interval, and the coding rate used for the inner coder. The useful transmission capacity U is given by

$$U = \frac{\nu}{T} \frac{1512}{2048} \frac{1}{(1 + \Delta/T_U)} \frac{188}{204} C$$

where ν is the number of bits per symbol (2 for QPSK, 4 for 16-QAM, and 6 for 64-QAM); T is the elementary period (1/8 μ s for a 7 MHz channel, 7/48 μ s for a 6 MHz channel, and 7/64 μ s for an 8 MHz channel); Δ/T_U is the length of the guard interval; and C is the rate of the inner convolutional coder.

The value of U can be constructed as follows. The basic data rate of the DVB system is ν/T . This is reduced by the fact that not all carriers are used to carry data (1512/2048); some transmission capacity is used for the guard intervals $1/(1+\Delta/T_U)$, the overhead for the outer code (188/204), and the overhead for the inner code (C).

EXAMPLE 13.5—DVB Channel Capacity

Find the capacity of a 7-MHz DVB channel using QPSK modulation with a guard interval of one quarter of the symbol period and a 7/8-rate inner code.

A QPSK ($\nu = 2$), 7 MHz ($T = 1/8 \mu$ s) channel with guard interval $\Delta/T_U = 1/4$ and a 7/8-rate convolutional coder has a capacity of 7.62 Mbit/s. ■

13.5. CONCLUSION

Channel coding and modulation are an integral part of any digital television system. Although DVB and ATSC have developed different systems, both are based on a concatenated coder, employing a convolutional inner coder, a convolutional interleaver, and a Reed–Solomon outer coder. This is no coincidence, representing a very effective trade-off between performance and complexity, using the technology available at the time of development.

PROBLEMS

13.1 Fill in the blank cells in Table 13.9, relating the DVB channel capacity to the various parameters of the channel.

Table 13.9 DVB channel capacities for Q 13.1.

U (Mbit/s)	ν	Channel BW (MHz)	Δ/T_U	C
4.3544	2	7	1/4	1/2
15.834		7	1/32	3/4
	4	6	1/16	2/3
21.772	6	7		5/6
14.929	6		1/4	1/2

13.2 Complete the blank entries in Table 13.10. The notation is that used in Section 13.2.5.1.

Table 13.10 Reed–Solomon characteristics for Q 13.2.

n	k	Max(s)	Max(r)
255	251	4	
255	239		8
255	223	16	
	15	4	8
255		4	8

13.3 Using the data provided in Tables 13.4 and 13.6 and Section 13.4.1.5, confirm that each OFDM symbol uses 193 carriers for reference signals in the 2k mode and 769 carriers for reference signals in the 8k mode.

13.4 If a convolutional encoder of the type shown in Figure 13.21 is used to encode a data stream D into two streams X and Y , write down a logic equation that can be used to extract the stream D from

- (a) the current value of X and the previous values of D and
- (b) the current value of Y and the previous values of D .

13.5 Write down a logic equation to decode the output of the DVB inner convolutional coder and recover its input data in the error free case using

- (a) the current value of X and previous values of D and
- (b) the current value of Y and previous values of D .

13.6 A 5-register cyclic interleaver has input stream 10, 3, 4, 2, 8, 9, 3, 11, 19, 0, 5, 7, 2, 1, 12. Write down the output stream for start positions zero and four.

13.7 A 5×2 block interleaver has input stream 10, 3, 4, 2, 8, 9, 3, 11, 19, 0, 5, 7, 2, 1, 12, 14, 7, 5, 2, 1. Write down the output stream.

13.8 A 3-level convolutional interleaver with $n = 1$ has input stream 7, 2, 1, 7, 4, 2, 1, 12, 14, 10, 3, 4, 2, 8, 9, 3, 11, 19, 60, 5, followed by a continuous stream of value 17. All cells of the interleaver have the value zero prior to loading the first element of the input stream. Write down the output stream beginning from the time that the first element of the input stream appears at the output and ending when the element with value 5 is output.

Table 13.11 Relationship between input bit-stream rate, code rate, and output bit-stream rate for Q 13.9.

Input bit-stream rate	Code rate	Output bit-stream rate
100 kbit/s	1/2	200 kbit/s
	1/3	3 Mbit/s
300 kbit/s		450 kbit/s
150 bit/s	5/6	
	3/4	160 Mbit/s

- 13.9** Complete the missing entries in Table 13.11, relating the input data rate, code rate, and output bit-stream rate for a convolutional coder.
- 13.10** Figure 13.31 shows the VSB trellis encoder used for ATSC. Draw the block diagram of a decoder to reproduce X_1 and X_2 from R , assuming that no transmission errors occur.
- 13.11** Calculate the total bit rate of the DVB transmission parameter signaling in the 2k and 8k modes.
- 13.12** A DVB system, using the 8k mode, employs two transmitters on a single-frequency network. What is the maximum permissible difference in path length between a receiver and each of the transmitters before the difference in arrival times of the two transmissions exceeds the length of the guard interval? Perform the calculation for $\Delta/T_U = 1/4$ and $1/32$.
- 13.13** Calculate the average number of DVB transmission parameter signaling blocks transmitted each second on the carrier with index 34 in 2k and 8k modes.
- 13.14** Draw the block diagram of a decoder to recover the values of X_1 and X_2 from the sequence R generated by the encoder shown in Figure 13.31. This decoder need is only able to function correctly where all values of R are correctly received.
- 13.15** Calculate the number of data frames, data fields, data segments, and MPEG-2 transport packets per second carried by an ATSC digital television channel.
- 13.16** Calculate the proportions of the ATSC channel capacity used for carriage of
- (a) field sync,
 - (b) segment sync,
 - (c) trellis coder overhead,
 - (d) Reed–Solomon coder overhead, and
 - (e) MPEG-2 transport stream packets.
- 13.17** Calculate the proportions of the DVB channel capacity used for carriage of
- (a) TPS,
 - (b) scattered and continual pilots,
 - (c) convolutional coder overhead,
 - (d) Reed–Solomon coder overhead, and
 - (e) MPEG-2 transport stream packets.
- 13.18** Calculate the minimum bandwidth required to transmit a 64 kbit/s signal without significant intersymbol interference using
- (a) BPSK,
 - (b) QPSK,

- (c) 16-QAM, and
- (d) 64-QAM.

13.19 Calculate the minimum bandwidth (based on the Nyquist criterion) required to transmit a 128 kbit/s signal using

- (a) BPSK,
- (b) QPSK,
- (c) 16-QAM, and
- (d) 64-QAM.

13.20 Calculate the number of data bits carried by one DVB OFDM symbol in 2k and 8k modes for each of the three modulation schemes: QPSK, 16-QAM, and 64-QAM.

MATLAB EXERCISE 13.1

The aim of this exercise is to implement the DVB 8k mode using the inverse Fourier transform. The sequence of operations is to pack bits into modulation symbols, to interleave these modulation symbols across OFDM symbols, to convert this array of OFDM symbols to a sequence of samples to be sent to the transmitter, and finally to add the guard intervals.

1. Write a MATLAB function to convert an input array of bits into an output array of symbols for QPSK, 16-QAM, or 64-QAM.
2. Write a MATLAB function that takes as its input an array of QPSK, 16-QAM, or 64-QAM symbols, which it interleaves across 8k OFDM symbols using the parameters set out in Section 13.4.1.1, and returns an array of 8k OFDM symbols.
3. Implement an 8k OFDM modulator in accordance with the description in Section 13.4.1.1.
4. Add the guard interval to the beginning of each OFDM symbol.

MATLAB EXERCISE 13.2

The aim of this exercise is to evaluate the performance of the channel coding systems used in DVB and ATSC.

1. Write a MATLAB function that generates a random bit stream and convolutionally encodes this bit stream using the DVB convolutional encoder.
2. Write a MATLAB function that decodes this random bit stream using hard-decision Viterbi decoding.
3. Write a function that introduces random errors into the coded bit stream at a nominated error rate.
4. Grouping the output bits from the Viterbi decoder into 8-bit symbols, plot the output symbol error rate of the Viterbi decoder against its input bit error rate. This should be done for input bit error rates in the range 10^{-1} – 10^{-6} . You

may need to run these simulations a number of times and average the results to obtain useful plots.

5. Assuming that the input symbol errors to the DVB RS decoder are randomly spaced, use the information in the graph in Figure 13.19 to plot the output bit error rate of the DVB concatenated coder versus its input bit error rate.

Hint: the `convenc` and `vitdec` functions in the MATLAB communications toolbox provide convolutional encoding and decoding.

MATLAB EXERCISE 13.3

The aim of this exercise is to implement a convolutional interleaver in MATLAB.

1. Implement a 6-level convolutional interleaver using a MATLAB function. The argument to this function is an array of data to be interleaved. The delay D is one array element.
2. Extend the interleaver so that the number of levels can be specified as a second argument to the function.
3. Further extend the interleaver function so that the register size (expressed in array elements of the input data) can be specified as a third argument to the function.

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